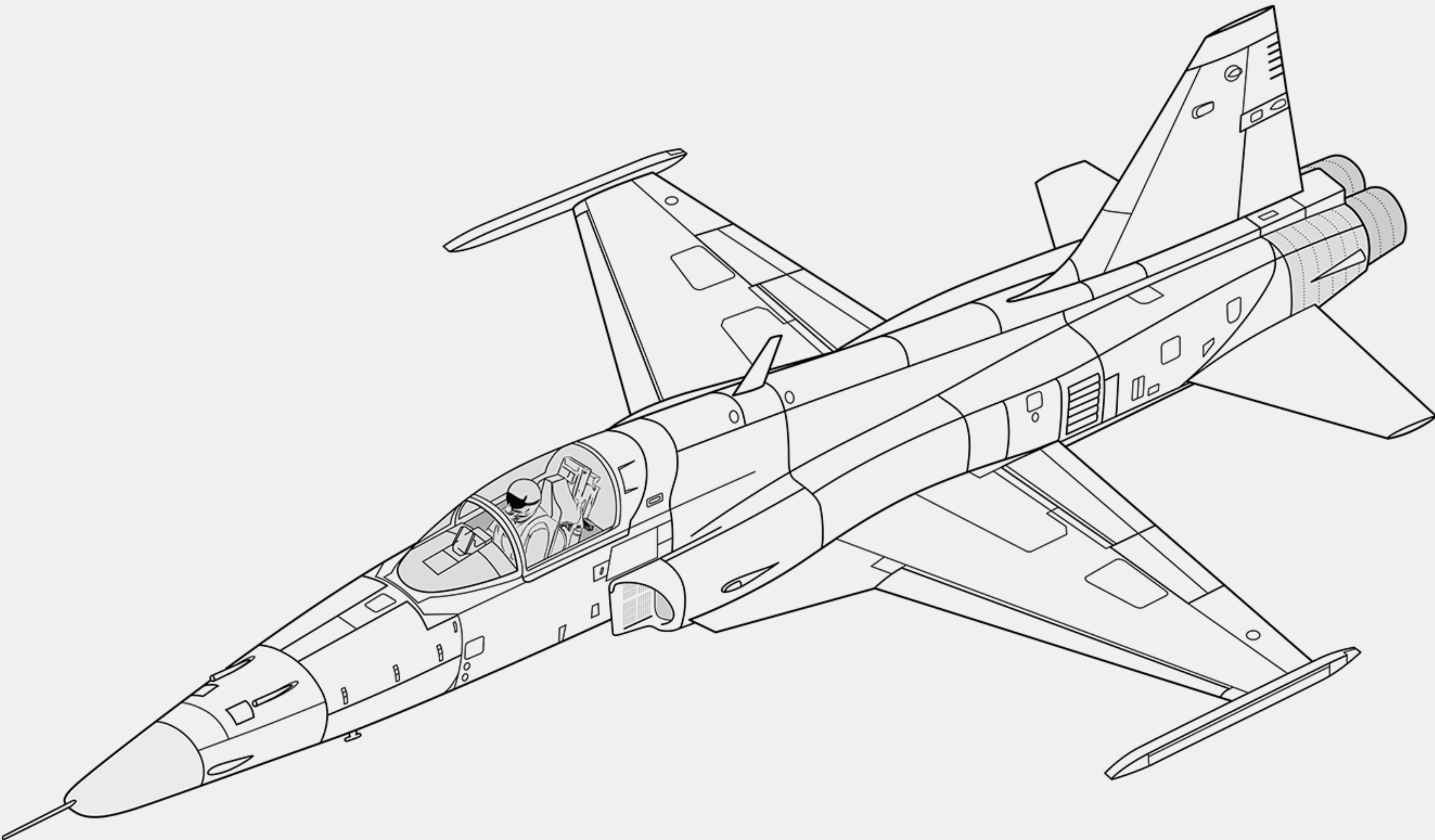


KNEEBOARD
SUITE

F-5E-3



Checklist-I 2

Checklist-II and Combat Checks 3

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Ordnance and Fuel Balancing 5

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Cockpit and Quickstart 9

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Cockpit Diagram (Large) 19

PREFLIGHT CHECK*

1. Ground Air Supply — Connect.
2. Pins and Covers — Check Removed.
3. Flap Lever — THUMB SW.
4. Speed Brake Switch — Neutral (OFF).
5. Flap Thumb Switch — UP.
6. Standby Attitude Indicator — Uncage (ccw).
7. Flight Instruments, Clock — Check and Set.
8. UHF, TACAN, NAV MODE — As Req.
9. Fuel Boost Pumps — LEFT and RIGHT.
10. Battery Switch — BATT.
11. Generators — L GEN and R GEN (rmb).
12. Fuel & Oxy Switch — GAGE TEST, QTY CHECK.
No indicator response indicates static inverter failure.
13. IFF / Master Switch — As Req / STBY.
14. Int and Ext Lights, Beacon — As Req.

STARTING ENGINES LEFT ENGINE

1. Ground Air Supply — Apply.
2. At 10% RPM, Start Button — PUSH.
3. Throttle — Advance to IDLE.
4. Engine RPM — Stabilized at 49–52%.
5. Hydraulic Pressure — Check 2800–3200 psi.
6. Generator Caution Light — Out.

GROUND AIR START RIGHT ENGINE

Omit this procedure if crossbleed start is to be used.

1. Same as for Left Engine.
2. Ground Air, Ground Power — Disconnect.

CROSSBLEED START RIGHT ENGINE

Recommended be used only in isolated areas.

1. Ground Air, Ground Power — Disconnect.
2. Wheel Chocks — Place.
3. Left Engine RPM — Increase to 95%.
4. Right Engine Start Button — PUSH.
5. At 10% RPM, Right Throttle — IDLE.
6. Left Throttle — Retard to Idle after Right Engine is stabilized at 49–52% RPM.
7. Generator Caution Lights — Out.

BEFORE TAXI

1. Speed Brake — In.
2. Flap Thumb Switch — AUTO (verify flaps at full).
3. Damper Switches — YAW and PITCH.
4. Pitch Trim — Set (refer to the table below).
5. Altimeter — To QNH (or to match field elevation).
6. Primary and Standby Attitude Indicators — Set 3 Degrees Nose Low.
7. Apply brakes; Wheel Chocks — Remove.

BEFORE TAKEOFF

For dehiked strut takeoff, expect up to 20% increase in V_R and up to 45% increase in takeoff roll.

1. Nose Strut Switch — EXTEND.
2. Optical Sight Mode Selector — As Req.
3. Radar Warning Receiver — As Req.
4. Radar Mode Selector — As Req.
5. Pitot Heat, Engine Anti-Ice — As Req.
6. HSI/ IFF — To runway course / As Req.
7. Canopy — Closed.
8. Caution and Warning Lights — Out.
9. Ext Lights, Beacon — As Req.

TAKEOFF

Do not engage nosewheel steering above 65 knots.

1. Apply brakes. Set MIL power RPM.
2. Release brakes. MAX thrust — As Req.
3. Aft stick — At 10 knots below liftoff speed.

WEIGHT ×1000 lbs	TAKEOFF DATA		LIFTOFF SPEED	PITCH TRIM	
	LOADOUT				
≤ 15	CLEAN, NO AMMO	143-145	6		
15,5-16	G M	153-155	7		
17-18	G M 1×F	164-168	7		
19	G M 3×F ¹⁵⁰	166-168	7		
19-21	G M 1×F B R	168-175	8		
22	G M 3×F	178-180	8		
≥ 23	G M B R	185-190	9		

* Kneeboard adjustments require cold and dark aircraft with battery off. Ground crew requires either an open canopy, battery power, or ground power.

If gun ammo is removed from loadout, decrease the liftoff speed by 10 kts.

G)un ammo
M)issiles
F)uel tanks
B)ombs
R)ockets

AFTER TAKEOFF. CLIMB.

1. Gear — Up.
2. Flap Thumb Switch — As Req.
3. External Fuel/Autobalance — As Req.
4. Oxygen, Cockpit Pressurization — Check.
5. Altimeter — As Req.

DESCENT

1. Armament Safety Check — Complete.
2. Canopy Defog, Engine Anti-Ice, and Pitot Heat — As Req.
3. Oxygen — Check.
4. Altimeter — Check and Set.
5. Ext and Landing Lights — As Req.

BEFORE LANDING

1. Altimeter — Check and Set.
2. Manual Crossfeed — Discontinue.
3. Hydraulic Pressure — 2800–3200 psi.
4. Flap Thumb Switch — AUTO.
5. Gear — Down (below 260 kts).
6. AOA — On Speed.

If AOA is inop, maintain 145 kts + weight correction.
Add 5 kts for full guns ammo. Add 1 kts for each
200 lbs of fuel above 1000 lbs remaining.

Drag chute max deployment speed: 180 kts.

GO-AROUND

1. *Throttles — MIL (MAX if necessary).*
2. *Speed Brake — In.*
3. *Confirm positive rate of climb; Gear — Up.*
4. *Flaps — As Req.*

ARMAMENT SAFETY CHECK

0. Fuel Quantity — Check.
 1. Guns, Msl & Camr Switch — OFF.
 2. Bombs Arm Switch — SAFE.
 3. Armament Select Switches — OFF.
 4. External Stores Selector — SAFE.
 5. Sight Mode Selector — OFF.
 6. Missile Volume — Off (ccw).

COMPLETE AS SOON AS PRACTICABLE
AFTER WEAPONS EMPLOYMENT

AFTER LANDING

1. Drag Chute — Jettison (if deployed).
Prior to opening canopy:
2. Check Cabin Pressure Altimeter;
Cockpit Pressurization Switch — RAM DUMP.
3. Flap Thumb Switch — UP.
4. Speed Brake — Out.
5. Optical Sight, RWR, and Radar — OFF.
6. Pitot Heat and Engine Anti-Ice — OFF.
7. Ext and Landing Lights, Beacon — As Req.
8. IFF — OFF.
9. Pitch Trim — Reset (6 to 7 increments).

ENGINE SHUTDOWN

1. Canopy — Open before shutdown.
2. Cockpit Pressurization Switch — NORMAL.
3. Apply brakes; Wheel Chocks — Place.
4. All Unguarded Switches (except Battery, Generators, and Fuel Boost Pumps) — OFF.
5. Seat — Full Up.
6. Throttles — OFF.
7. Standby Attitude Indicator — Cage (pull + cw).
8. Battery Switch — OFF.

FLAP THUMB SWITCH

UP: Max range. Max endurance without stores.

FIXED: Max endurance with stores.

AUTO: Max turn performance. For all phases of
maneuvering flight. Not for enroute cruise.



COMBAT CHECKS

14	F-14A/GR/B	123	88	21	MiG-21Bis	12	09	37	AJS-37 Viggen	2	06	JF	JF-17	23	46
15	F-15C	123	49	23	MiG-23MLD	12	14	50	A-50 / KJ-2000		AWACS	M2	M-2000C	12	26
	F-15E	23	49	25	MiG-25PD	12	25	64	AH-64D BLK.II	3	05		Mirage 2000-5	123	24
	F-15E S4+	123	49	29	MiG-29A/G	12	38		(with or w/o MMA)			U	A-6E	2	04
16	F-16A	123	38		MiG-29S	123	33	E2	E-2D		AWACS		AH-64D (old)	3	05
	F-16A MLU	23	38		Su-27/33	12	32	E3	E-3A		AWACS		C-101CC	2	04
	F-16C/CM	23	38		J-11A	123	32	F1	Mirage F1	12	22		H-6J		BOMBER
18	F/A-18A	12	27	30	Su-30	123	37	F2	Tornado GR4	2	04		Tornado IDS	2	04
	F/A-18C	123	42	31	MiG-31	12	37	F4	F-4E-45MC	12	27		Tu-95MS		BOMBER
19	MiG-19P	2	03	34	Su-34	123	35	F5	F-5E/E-3/FC	2	04		SH-60B		HELO

SURFACE												NAVAL			
S	Nebo 1L13 / Nebo 55G6 / AN/FPS-117											15	Project 22160 Tor-M2 CORVETTE	C	10/39
	Flat Face B	[2-3-5]										40	LHA-1 Tarawa CV	F	08/13
BB	Big Bird	[10]										49	Oliver Hazard Perry FRIGATE	C	26/++
CS	Clam Shell	[10]										AE	Arleigh Burke DESTROYER	C	80/++
DE	Dog Ear Sborka	[13-15-19]											Ticonderoga CRUISER	C	37/++
FC	Fire Can SON-9											FR	CV-59 Forrestal CV	C	10/24
HQ	HQ-7B	[7]										HN	Type 052C DESTROYER	C	90/++
SD	Snow Drift	[11]											Pyotr Velikiy CRUISER	C	86/++
TS	Tin Shield	[5-10]										HP	Grisha (2025) CORVETTE	C	07/17
U	Freya / Würzburg												Grisha (old) TO BE REMOVED	C	08/15
	Silkworm											IV	HMS Invincible CV	C	47/++
	Tall King	[5]										LC	La Combattante FAST ATTACK CRAFT	E	04/10
2	C 27/++ SA-2 Guideline S-75											MR	Type 052B DESTROYER	C	31/++
													Type 054A FRIGATE	C	33/++
3	C 13/++ SA-3 Goa S-125											PS	Type 071 Amphib.dock TRANSPORT	E	06/10
5	C 160/+ SA-5 Gammon S-200												Molniya CORVETTE	E	03/09
6	C 18/27 SA-6 Gainful Kub											SC	LS Ropucha BDK-775 TRANSPORT	E	06/09
7	E 10/18 HQ-7B											SS	CVN-71 Roosevelt CV		
8	C 07/17 SA-8 Gecko Osa												CVN-72 Lincoln CV		
10	C 48/++ SA-10 Grumble S-300PS												CVN-73 Washington cv	C	10/24
11	C 29/++ SA-11 Gadfly Buk												CVN-74 Stennis CV	F	08/13
13	F 03/12 SA-13 Gopher Strela-10M												CVN-75 Truman cv		
15	C 09/20 SA-15 Gauntlet Tor											SW	Kuznetsov 2017 cv	C	08/20
	C 10/39 SA-15 Gauntlet M2 Tor-M2												Kuznetsov (old) cv	C	08/20
19	E 05/12 SA-19 Grison Tunguska											T2	Moskva CRUISER	C	48/++
22	C 11/33 SA-22 Greyhound Pantsir-S1												Project 22160 CORVETTE	E	04/13
A	E 03/10 Gepard											TP	Condell-class FRIGATE	E	02/07
	E 02/06 Vulcan												CNS Almirante Condell / CNS Almirante Lynch		
	E 02/07 ZSU-23-4 Shilka												Leander-class FRIGATE	E	06/11
CR	E 02/05 LPWS Centurion Phalanx												HMS Achilles / HMS Andromeda / HMS Ariadne		
HK	C 14/++ Hawk												Neustrashimy FRIGATE	C	08/20
IT	F 22/++ IRIS-T SLM												Rezky FRIGATE	C	08/15
NS	C 13/35 NASAMS AIM-120B											U	ARA Veinticinco de Mayo CV	E	02/10
	C 13/40 NASAMS AIM-120C												Essex Class 1944 CV	E	02/07
P	C 60/++ Patriot												Type 021-1 MISSILE BOAT	E	02/12
RO	C 05/20 Roland														
RP	E 05/10 Rapier with Blindfire Radar														
RT	E 05/10 Rapier with or without Tracker														
U	E 10/20 HQ-7 'Player' AI- or user-controlled														

1 — SARH, 2 — IR, 3 — ARH — SUPPORTED A-A MISSILES, FOLLOWED BY THE MAX LAUNCH RANGE AT 40.000 ft, nm.
18/27 — SAM'S MAX FIRING RANGE (nm) AND ALTITUDE (×1000 ft ABOVE GROUND LEVEL; ++ WHEN EXCEEDS 50.000 ft).
CM TO USE: C — CHAFF, F — FLARES, E — EVADE. [THE SAM] TO WHICH THIS RADAR IS USUALLY LINKED.
ALL RANGES WERE MEASURED USING THE F/A-18C. F-5's RWR DOES NOT DISTINGUISH AIRBORNE AND SURFACE EMITTERS.

○ TRACKING (FLASHING — GUIDING THE MISSILE) ◇ PRIMARY THREAT M MISSILE

up to 4510

up to 395

0

Internal fuel 1970+2540 lbs

Two 20mm M39-A3 cannons

2x280 HEI, CM, AP, or TP rounds

Up to 60 chaff, up to 30 flares;

program via kneeboard or in ME

7

6

5

4

3

2

1

127

121

161

343

Outboard pylon

Inboard pylon

Centerline pylon

Multiple Ejector Rack

TANKS, PODS

1123

2010

662

139

205

150 gal fuel tank 970 lbs

275 gal fuel tank 1770 lbs

MXU-648 travel pod

AN/ASQ-T50 ACMI pod

Smokewinder

AA MISSILES

~180

190

AIM-9B / J / P / P3 / P5

Captive AIM-9M

ROCKETS

Program via kneeboard or in ME:

512

518

461

459

512

468

408

410

254

256

267

254

234

236

234

254

238

216

216

19x Hydra M151 HE soft targets

19x Hydra M156 SM white smoke

19x Hydra Mk 5 HEAT anti-armor

19x Hydra Mk 61 TP practice

19x Hydra WTU-1/B TP practice

19x FFAR M156 SM white smoke

19x FFAR Mk 1 HE soft targets

19x FFAR Mk 5 HEAT anti-armor

7x Hydra M151 HE soft targets

7x Hydra M156 SM white smoke

7x Hydra M257 IL parach.flare

7x Hydra M274 TP-SM practice

7x Hydra Mk 1 HE soft targets

7x Hydra Mk 5 HEAT anti-armor

7x Hydra Mk 61 TP practice

7x Hydra WTU-1/B TP practice

7x FFAR M156 SM white smoke

7x FFAR Mk 1 HE soft targets

7x FFAR Mk 5 HEAT anti-armor

LAU-3

LAU-68

Empty weight:

10660 lbs

MTOW:

24660 lbs

NOSE: N NOSE PLUGS		TAIL: T TAIL PLUGS		Underscored: Default L: LASER CODE A: COATING
1	M904E4 mech	3	M905 mech	
2	DSU-33 prox	4	FMU-139 elec	
		5	FMU-143 elec penetr	
		6	FMU-152 JPF	

BOMBS

287

785

750

503

2515

552

2760

1000

1970

611

1131

2115

512

512

618

25

8x LUU-2 parach.flare

CBU-52B cluster, soft targets

M117 750 lbs

1x Mk-82 500 lbs

5x Mk-82 req. MER

1x Mk-82 Snakeye 500 lbs, retrd

5x Mk-82 Snakeye req. MER

Mk-83 1000 lbs

Mk-84 2000 lbs

GBU-12 500 lbs

GBU-16 1000 lbs

GBU-10 2000 lbs

BDU-50HD 500 lbs, ballute

BDU-50LD 500 lbs

BDU-50LGB 500 lbs, laser

BDU-33

6

5

4

3

2

287

785

750

503

2515

552

2760

1000

1970

611

1131

2115

512

512

618

25

8x LUU-2 parach.flare

CBU-52B cluster, soft targets

M117 750 lbs

1x Mk-82 500 lbs

5x Mk-82 req. MER

1x Mk-82 Snakeye 500 lbs, retrd

5x Mk-82 Snakeye req. MER

Mk-83 1000 lbs

Mk-84 2000 lbs

GBU-12 500 lbs

GBU-16 1000 lbs

GBU-10 2000 lbs

BDU-50HD 500 lbs, ballute

BDU-50LD 500 lbs

BDU-50LGB 500 lbs, laser

BDU-33

1 2 N 3 4 6 T A

4 5 6 L A

1 2 N 3 4 6 T

1 2 N 3 4 6 T

L

WEIGHT WITHOUT PYLONS AND RACKS, LBS

FUEL BALANCING

The right fuel system carry 570 lbs more fuel than the left system. This requires in-flight balancing on a fully fueled aircraft.

When external tanks are carried, use inboard tanks first, centerline tank next, and internal fuel last.

Begin transfer or balancing as soon as practicable after takeoff.

Check the fuel quantity gage operation before proceeding with either procedure.

EXTERNAL FUEL TRANSFER

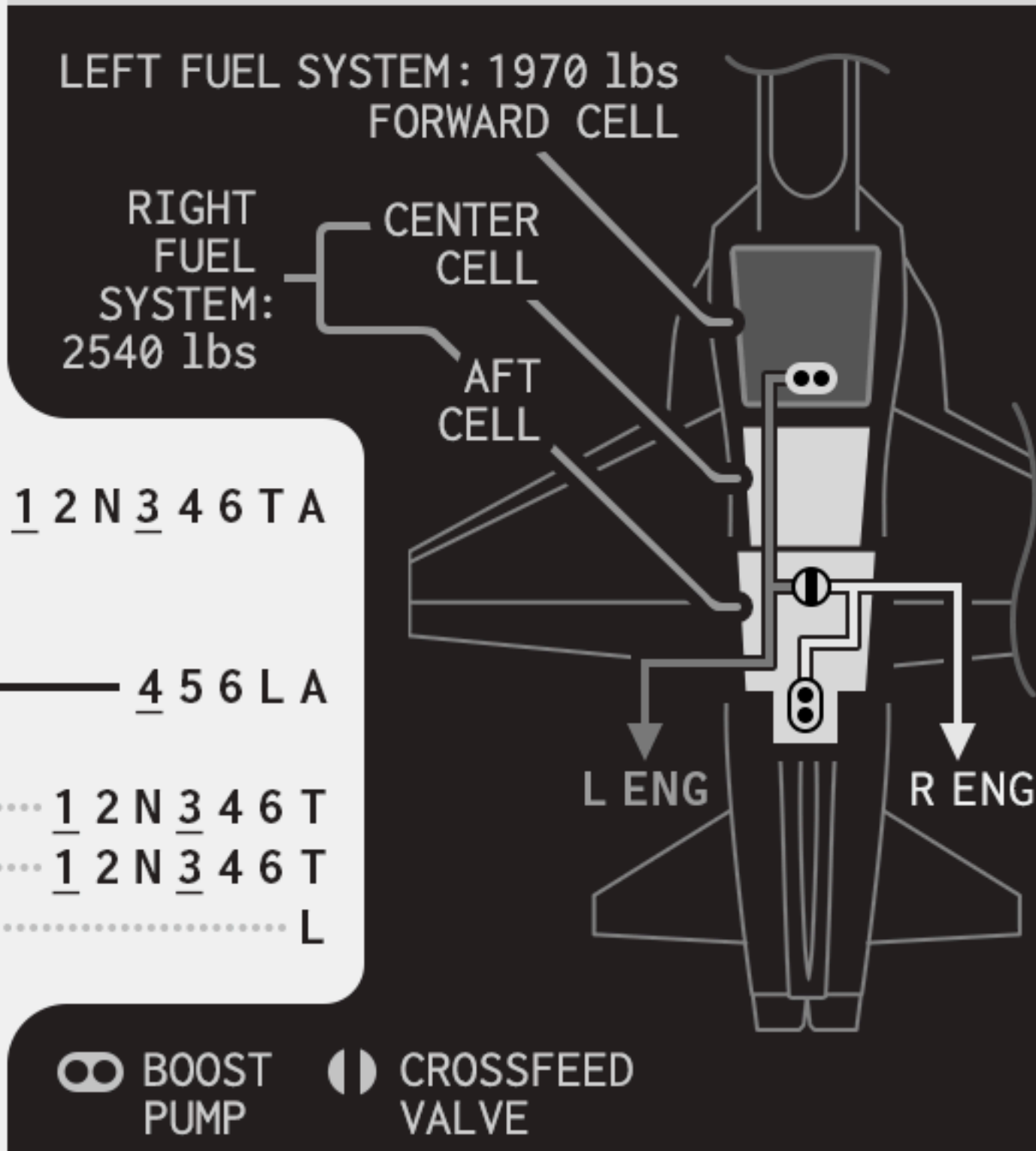
- Transfer from ext tanks inhibits balancing.
- (If carried) Ext Fuel PYLONS Switch — On.
 - When completed (indicated by the EXT TANKS EMPTY caution light); PYLONS Switch — OFF.
 - (If carried) Ext Fuel CL Switch — On.
 - When completed; CL Switch — OFF.

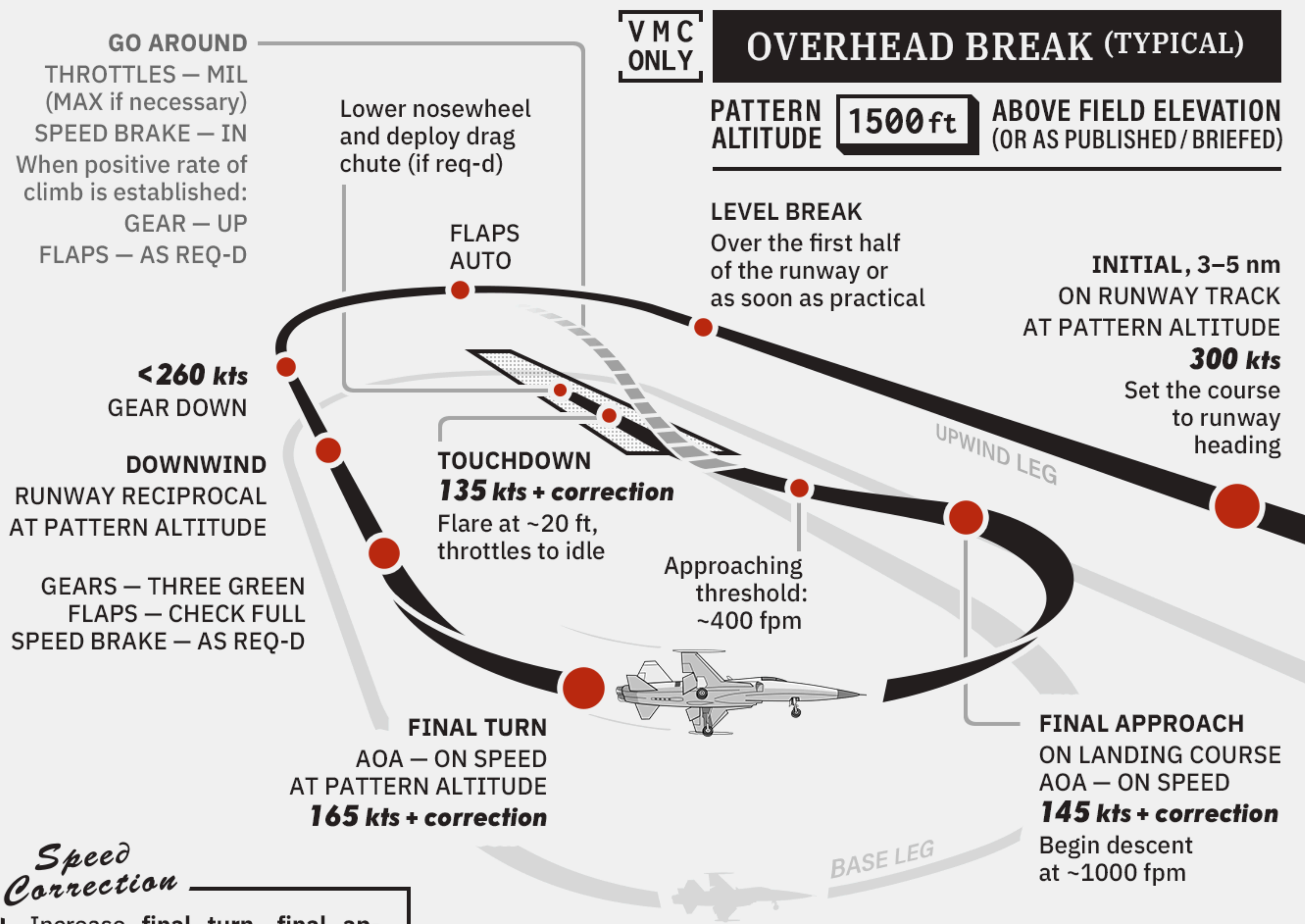
AUTOBALANCING

- Auto Balance Switch automatically returns to center position when system is balanced.
- Crossfeed Switch — OFF.
 - Boost Pumps — ON.
 - Auto Balance Sw. — To low fuel side.

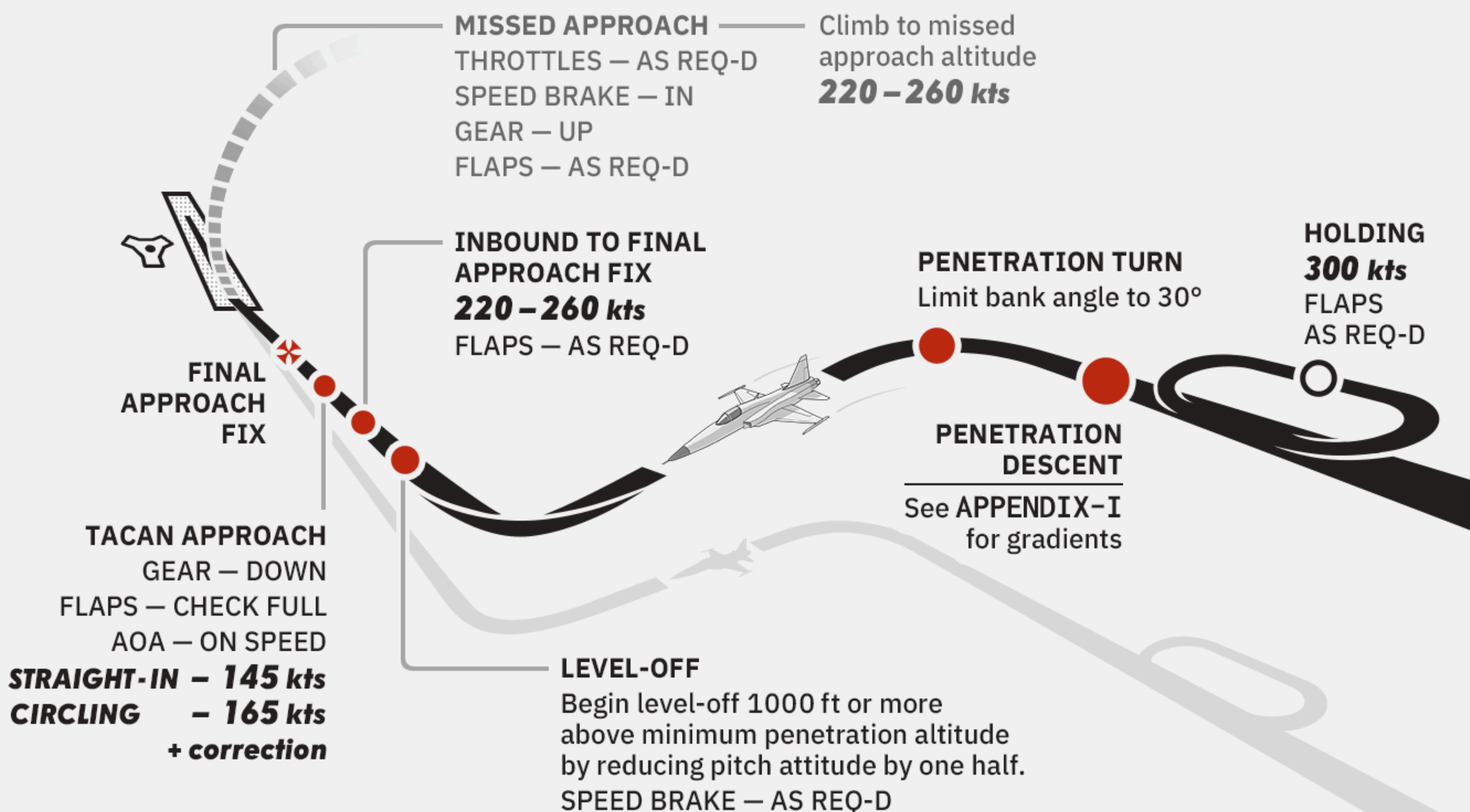
MANUAL BALANCING

- Crossfeed Switch — CROSSFEED.
- Boost Pump (on low fuel side) — OFF.
- System Balanced; Disabled Boost Pump — On.
- After at least 2 min; Crossfeed — OFF.





TACAN APPROACH (TYPICAL)



PENETRATION DESCENT GRADIENTS*

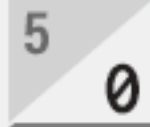
KLAS TO MAINTAIN	300	270	300	270	250	250
RPM TO MAINTAIN	80%	IDLE	80%	IDLE	IDLE	80%
FLAPS	FXD	UP	FXD	FXD	FXD	FXD
SPEED BRAKE	IN	IN	OUT	OUT	IN	OUT
REFERENCE VS, fpm	2500	5000	4500	6000+	3500	3000
DISTANCE FOR EACH 1000 feet	2-3 nm	1.2 nm	1.3 nm	0.9 nm	1.3 nm	2.0 nm

RECOMMENDED DESCENT RATES ON THE GLIDE PATH				
FUEL, lbs	< 3700		> 3700	
X-WIND	No	Yes	No	Yes
VS, fpm	600	400	400	300

MAXIMUM CROSSWIND COMPONENT DURING LANDING		
GROSS WEIGHT, lbs	W/DRAG CHUTE	W/O DRAG CHUTE
BELOW 15,500	20 kts	35 kts
ABOVE 15,500	25 kts	35 kts

* Conditions:
GW 14,000 lbs,
2000 lbs of fuel,
no stores,
no wind.
Distances vary
with drag index,
ground speed,
and wind.

HEADWIND
COMPONENT



CROSSWIND
COMPONENT

WIND COMPONENTS

Angle between wind direction and runway heading										
	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°
5	5 0	5 1	5 2	5 3	4 4	3 4	3 5	2 5	1 5	0 5
10	10 0	10 2	9 3	9 5	8 6	6 8	5 9	3 9	2 10	0 10
15	15 0	15 3	14 5	12 7	11 10	9 12	7 13	5 14	3 15	0 15
20	20 0	20 3	19 7	17 10	15 13	13 15	10 17	7 19	3 20	0 20
25	25 0	25 4	24 9	21 13	19 16	16 19	13 22	9 24	4 25	0 25
30	30 0	30 5	28 10	26 15	23 19	19 23	15 26	10 28	5 30	0 30
35	35 0	35 6	33 12	30 18	26 23	23 27	18 30	12 33	6 35	0 35
40	40 0	39 7	38 14	35 20	31 26	26 31	20 35	14 38	7 39	0 40
45	45 0	44 8	42 15	39 23	35 29	28 35	23 39	15 43	8 45	0 45
50	50 0	49 9	47 17	43 25	38 32	32 38	25 43	17 47	9 49	0 50
55	55 0	54 9	52 19	47 28	42 35	35 43	28 48	19 52	9 54	0 55
60	60 0	59 10	56 21	52 30	46 39	39 46	30 52	21 56	10 59	0 60
65	65 0	64 11	61 22	56 33	50 42	42 50	33 57	23 62	11 64	0 65
70	70 0	69 12	66 24	61 35	54 45	45 54	35 61	24 66	12 69	0 70

WIND SPEED, kts

IMPROMPTU VOR NAVIGATION

TACAN DME/VOR PAIRING TABLE		109.30	30X	110.75	44Y	112.20	59X	113.65	83Y	115.10	98X	116.55	112Y
		109.35	30Y	110.80	45X	112.25	59Y	113.70	84X	115.15	98Y	116.60	113X
		109.40	31X	110.85	45Y	112.30	70X	113.75	84Y	115.20	99X	116.65	113Y
108.00	17X	109.45	31Y	110.90	46X	112.35	70Y	113.80	85X	115.25	99Y	116.70	114X
108.05	17Y	109.50	32X	110.95	46Y	112.40	71X	113.85	85Y	115.30	100X	116.75	114Y
108.10	18X	109.55	32Y	111.00	47X	112.45	71Y	113.90	86X	115.35	100Y	116.80	115X
108.15	18Y	109.60	33X	111.05	47Y	112.50	72X	113.95	86Y	115.40	101X	116.85	115Y
108.20	19X	109.65	33Y	111.10	48X	112.55	72Y	114.00	87X	115.45	101Y	116.90	116X
108.25	19Y	109.70	34X	111.15	48Y	112.60	73X	114.05	87Y	115.50	102X	116.95	116Y
108.30	20X	109.75	34Y	111.20	49X	112.65	73Y	114.10	88X	115.55	102Y	117.00	117X
108.35	20Y	109.80	35X	111.25	49Y	112.70	74X	114.15	88Y	115.60	103X	117.05	117Y
108.40	21X	109.85	35Y	111.30	50X	112.75	74Y	114.20	89X	115.65	103Y	117.10	118X
108.45	21Y	109.90	36X	111.35	50Y	112.80	75X	114.25	89Y	115.70	104X	117.15	118Y
108.50	22X	109.95	36Y	111.40	51X	112.85	75Y	114.30	90X	115.75	104Y	117.20	119X
108.55	22Y	110.00	37X	111.45	51Y	112.90	76X	114.35	90Y	115.80	105X	117.25	119Y
108.60	23X	110.05	37Y	111.50	52X	112.95	76Y	114.40	91X	115.85	105Y	117.30	120X
108.65	23Y	110.10	38X	111.55	52Y	113.00	77X	114.45	91Y	115.90	106X	117.35	120Y
108.70	24X	110.15	38Y	111.60	53X	113.05	77Y	114.50	92X	115.95	106Y	117.40	121X
108.75	24Y	110.20	39X	111.65	53Y	113.10	78X	114.55	92Y	116.00	107X	117.45	121Y
108.80	25X	110.25	39Y	111.70	54X	113.15	78Y	114.60	93X	116.05	107Y	117.50	122X
108.85	25Y	110.30	40X	111.75	54Y	113.20	79X	114.65	93Y	116.10	108X	117.55	122Y
108.90	26X	110.35	40Y	111.80	55X	113.25	79Y	114.70	94X	116.15	108Y	117.60	123X
108.95	26Y	110.40	41X	111.85	55Y	113.30	80X	114.75	94Y	116.20	109X	117.65	123Y
109.00	27X	110.45	41Y	111.90	56X	113.35	80Y	114.80	95X	116.25	109Y	117.70	124X
109.05	27Y	110.50	42X	111.95	56Y	113.40	81X	114.85	95Y	116.30	110X	117.75	124Y
109.10	28X	110.55	42Y	112.00	57X	113.45	81Y	114.90	96X	116.35	110Y	117.80	125X
109.15	28Y	110.60	43X	112.05	57Y	113.50	82X	114.95	96Y	116.40	111X	117.85	125Y
109.20	29X	110.65	43Y	112.10	58X	113.55	82Y	115.00	97X	116.45	111Y	117.90	126X
109.25	29Y	110.70	44X	112.15	58Y	113.60	83X	115.05	97Y	116.50	112X	117.95	126Y

Set the UHF radio to ADF mode and tune it to a VOR frequency by using T instead of the leading 1. Set the TACAN to T/R mode and tune it to a VORDME channel equivalent from the table above.

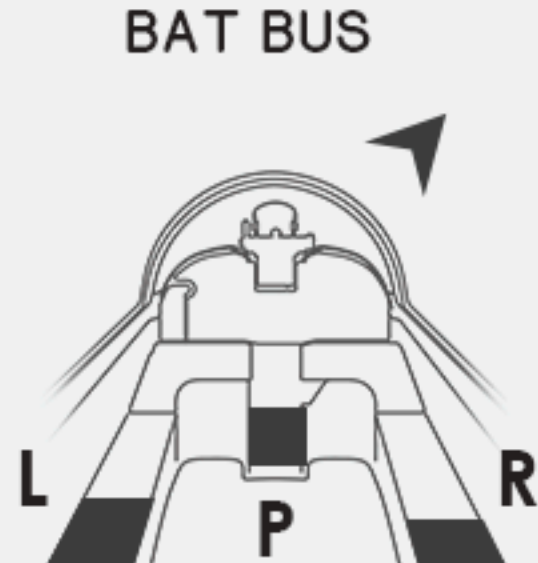
Switch between the DF and TACAN NAV MODES as desired to display bearing or distance to the VORDME beacon on the HSI.

Switch the UHF back to BOTH whenever needed to talk to ATC.

CIRCUIT BREAKER PANELS

ARMAMENT				115/200 VOLT AC				28 VOLT DC			
LEFT AIM-9 POWER (5)	LEFT GUN FIRING (5)	28 VOLT AC POWER (1½)	ATTD & HDG REF SYS (5)	CENTRAL AIR DATA COMPUTER (5)	ENG IGN L ENG INST & HYD IND (5)	PYLON TANK FUEL CONT (5)	UHF CMND RADIO (5)	L BOOST CL & TIP TANK FUEL CONT (5)	LEFT LE FLAP CONT (5)	IGNITION INVERTER POWER (10)	RIGHT LE FLAP CONT (5)
RIGHT AIM-9 POWER (5)	RIGHT GUN FIRING (5)	TRIM CONTROL (5)	ATTD & HDG REF SYS (5)	TRIM CONTROL (5)	L ENG AUX DOOR (5)	L ENG START & AB CONT (5)	LEFT TE FLAP CONT (5)	R ENG START & AB CONT (5)	RIGHT TE FLAP CONT & IND (5)		
CABIN COND (5)	FLT DIR COMPUTER	FUEL QTY - PRIMARY (5)	ATTD & HDG REF SYS (5)	TACAN (5)	DIRECTIONAL GYRO						

LEFT				PEDESTAL				RIGHT			
28 VOLT DC				115/200 VOLT AC							
WPN PWR LEFT INBD (7½)	WPN PWR CENTER LINE (7½)	WPN PWR RIGHT INBD (7½)		PITOT HEATER (7½)		R OIL & HYD IND FUEL QTY SEL (1)	CABIN AIR VALVES (5)				
WPN PWR LEFT OUTBD (7½)	WPN ARMING (7½)	WPN PWR RIGHT OUTBD (7½)			INST LIGHTS (5)	R ENG AUX DOOR (5)	BLANKING ELECTRONIC UNIT				
JETTISON CONTROL (5)	WPN RELEASE (7½)	WPN MODE SEL & AIM-9-INTLK (7½)			CAUTION & WARN LIGHTS - DIM (5)	QXY QTY & CANOPY SEAL (5)	LDG - TAXI LAMP PWR (5)				
EMERGENCY ALL JETTISON (15)	LEFT AIM-9 CONT (7½)	RIGHT AIM-9 CONT (7½)									



NOTE ON RADIO AND ADF

While nominally limited to 225–399 MHz, the ‘ARC–164’ radio in DCS de-facto supports 100.000–399.975 MHz range.

Consequently, the ARA–50 in ADF + DF mode can display bearing to any signal in that range (voice, VOR, ILS, etc).

To tune to 1XX.XXX A frequencies, rotate the first radio dial knob to the T position.

LANDING LIGHTS OPERATION

If the position lights (the NAV knob) are on, the landing lights automatically extend and retract together with the landing gear.

The LDG & TAXI switch is used to turn the extended landing lights on and off.

Lights automatically switch to low intensity while on the ground.

Landing lights can stuck in the extended position if gear is retracted with position lights turned off. Cycle the NAV knob to resolve.

THE MODULE'S MISSING FEATURES

Consider the following when planning missions:

- Fire warnings, nozzle damage, and canopy cracks are purely cosmetic and have no impact on flying.
- Engine mechanical failures are not implemented.
- Battery cannot be discharged or overloaded.
- Icing is not simulated.
- SAI can operate on battery power indefinitely.
- ‘DC OVERLOAD’, ‘DIR GYRO’, ‘IFF’, and ‘INS’ cautions are not simulated.

The following **emergency procedures** are non-applicable and were not included in these kneeboards: ‘Nosewheel Shimmy’, ‘Electrical Fire’, ‘Smoke in Cockpit’, ‘Engine Malfunctions’, ‘Airframe Gearbox Failure’, ‘Ditching’.

The following **failure triggers** have no effect: ‘Fire Engine’, ‘Gearbox’, ‘Hydraulic System’, ‘Low Pressure Leakage’, ‘Rectifier’, ‘Nozzle Control System’, ‘Oil System’, ‘Oxygen System Total Failure’.

- PROHIBITED MANEUVERS**
- **Sustained 0-G or negative-G flight** at high engine power settings. May lead to engine fuel starvation and flameout within 10-120 seconds, depending on altitude and how much fuel flow exceeds 650 lb/h.
 - **Flying in icing conditions.** When icing conditions are anticipated, turn on the pitot heat, engine anti-ice, and canopy defog. Maintain at least 80% RPM to ensure effective engine anti-icing. Leave the area of icing conditions as soon as possible.
 - **Flying through thunderstorms or extreme turbulence.** If cannot be avoided, the best penetration speed is 300 kts. Trim for level flight. Do not change thrust and trim. Maintain attitude. Do not chase altitude and vertical velocity indicators.
 - **Exceeding aileron spring stop** except for spin recovery and emergencies. (*Structural damage is not simulated.*)

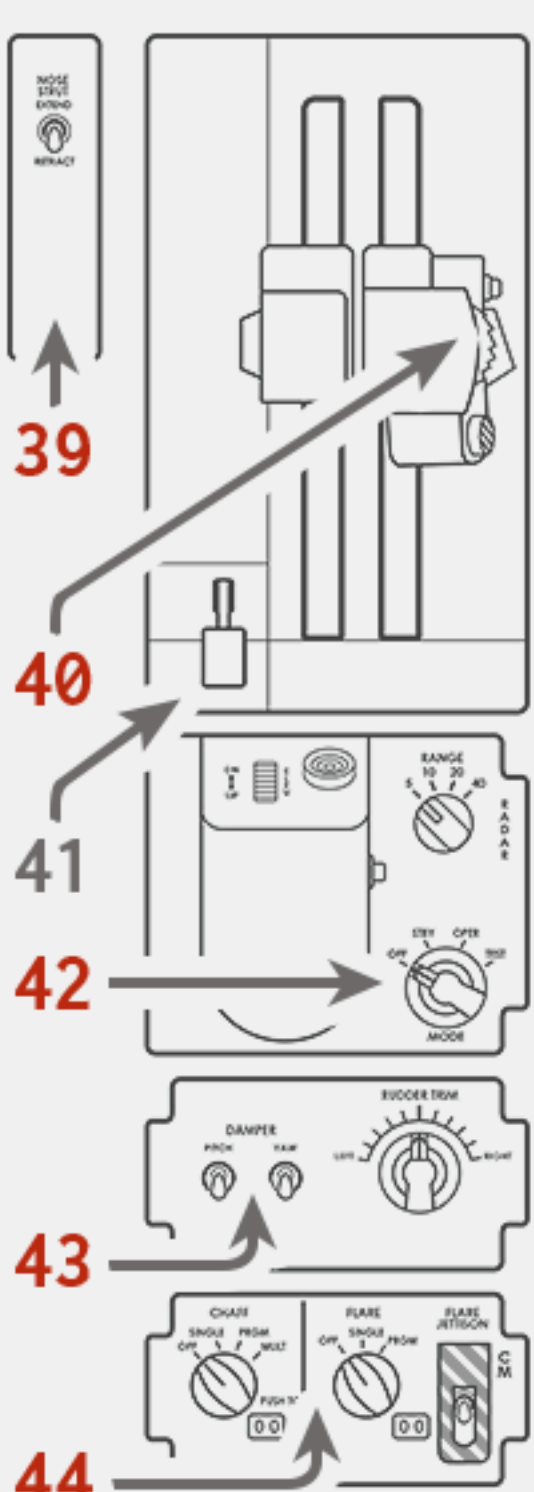
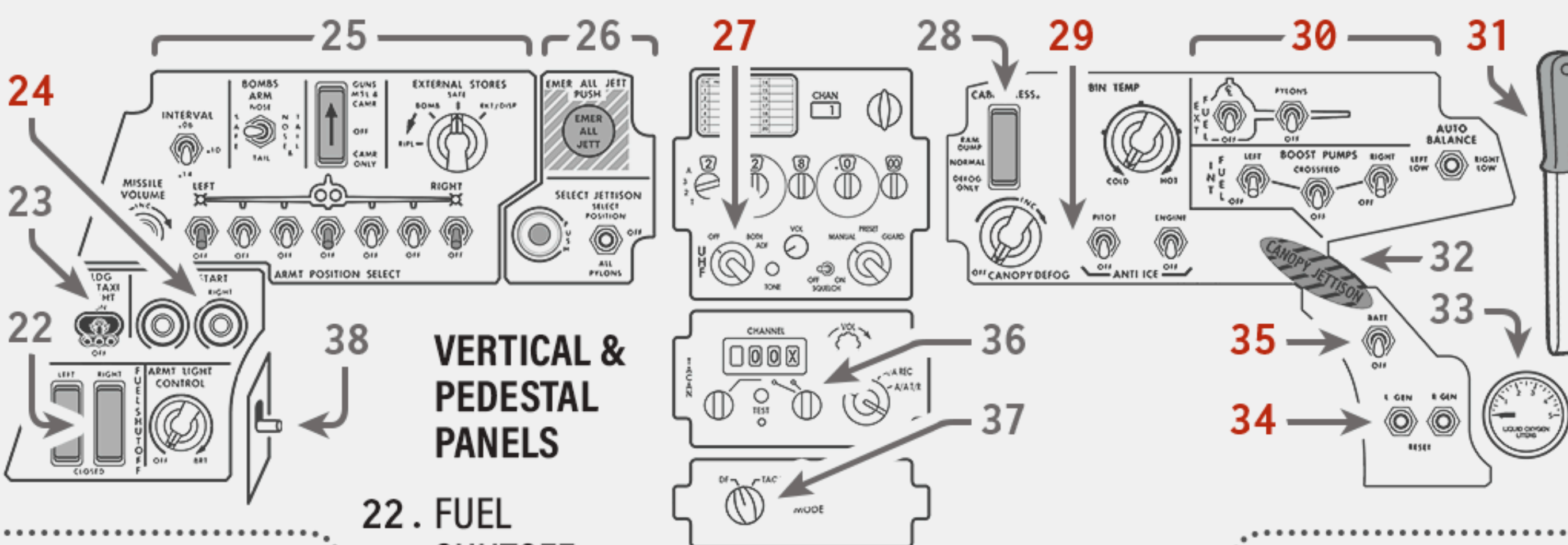
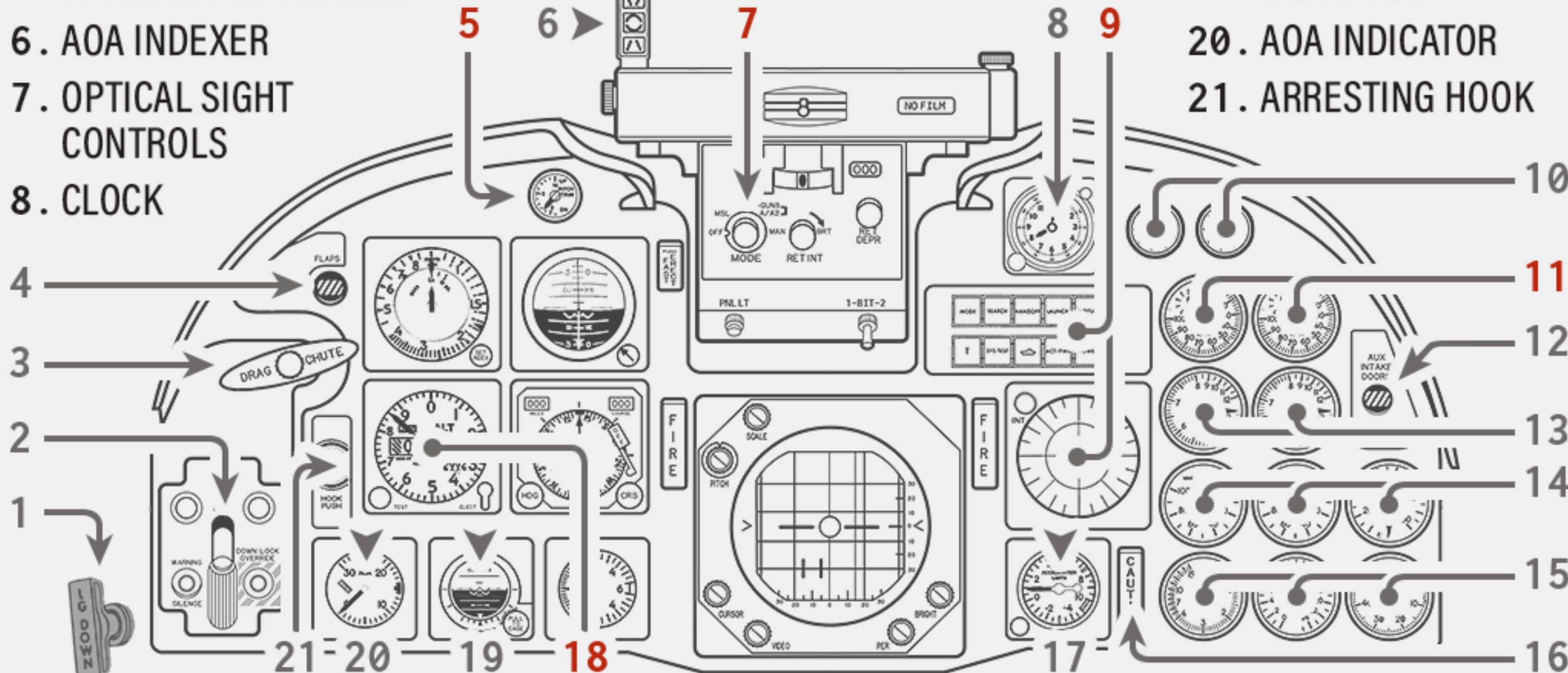
INSTRUMENT PANEL

QUICKSTART

1. LANDING GEAR ALTERNATE RELEASE HANDLE
2. LANDING GEAR LEVER & WARNING SILENCE BUTTON
3. DRAG CHUTE T-HANDLE
4. FLAP INDICATOR
5. PITCH TRIM INDICATOR
UNDER THE GLARESHIELD

9. RADAR WARNING RECEIVER
10. HYDRAULIC PRESSURE INDICATORS
11. ENGINE TACHOMETERS
12. AUX INTAKE DOORS INDICATOR
13. EGT INDICATORS
14. NOZZLE POSITION & OIL PRESSURE INDICATORS

15. FUEL FLOW & FUEL QTY INDICATORS, CABIN PRESSURE
16. MASTER CAUTION
17. ACCELEROMETER
18. ALTIMETER
19. STANDBY ATTITUDE INDICATOR
20. AOA INDICATOR
21. ARRESTING HOOK



22. FUEL SHUTOFF SWITCHES
23. LANDING & TAXI LIGHT SWITCH
24. ENGINE START BUTTONS
25. ARMAMENT PANEL: MISSILE VOLUME
26. JETTISON CONTROLS
27. UHF RADIO/ADF CONTROLS
28. COCKPIT PRESSURIZATION SWITCH
29. PITOT & ENGINE ANTI-ICE SWITCHES, CANOPY DEFOG

30. BOOST PUMP, CROSSFEED, FUEL TRANSFER & AUTO BALANCE SWITCHES
31. CANOPY HANDLE
32. CANOPY JETTISON
33. OXYGEN QUANTITY
34. GENERATOR SWITCHES
35. BATTERY SWITCH
36. TACAN CONTROLS
37. NAVIGATION MODE
38. GEAR ALTERNATE RELEASE RESET

LEFT & RIGHT CONSOLES

39. NOSE STRUT SWITCH
40. RIGHT THROTTLE: FLAP THUMB & SPEED BRAKE SWITCHES
41. FLAP LEVER
42. RADAR MODE SWITCH

43. STABILITY AUGMENTER CONTROLS: PITCH & YAW DAMPER SWITCHES
44. COUNTER-MEASURES DISPENSER CONTROLS: CHAFF & FLARES
45. OXYGEN REGULATOR

46. CAUTION LIGHTS
47. IFF/SIF CONTROLS
48. FUEL & OXY GAGE TEST / QUANTITY CHECK SWITCH
49. LIGHTING CONTROLS: BEACON & WARNING LIGHTS TEST SWITCHES

- 27 UHF BOTH (*rmb*)
- 30 BOOST PUMPS LEFT & RIGHT
- 35 BATT
- 34 L GEN & R GEN (*rmb*)
- Ground Air Supply: Connect & Apply
- 11 At >10% Left RPM:
- 24 ENGINE START LEFT
- Left Throttle: IDLE (*RAIt + Home*)
- 11 Wait for Left RPM to stabilize at ~50% (in about 30 sec)
- Ground Air Supply: Apply
- 11 At >10% Right RPM:
- 24 ENGINE START RIGHT
- Right Throttle: IDLE (*RShift + Home*)
- 11 Wait for Right RPM to stabilize at ~50%
- Ground Air Supply: Disconnect

- 31 Close (*LCtrl+C*)
- 40 Speed Brake IN (*fwd*) (*rmb*)
- 40 Flap Thumb Switch AUTO (*aft*)
- 43 PITCH & YAW
- 18 Set QNH
- 5 Set Pitch Trim based on loadout and note rotation speed for takeoff:
- CLEAN: 7 / 150 kts
- MEDIUM: 8 / 165 kts
- HEAVY: 9 / 180 kts

BEFORE TAKEOFF:

- 39 Extend Strut (*fwd*)
- 7 Sight as desired
- 9 RWR POWER
- 42 Radar OPER
- 44 CMs as desired
- 29 PITOT

GENERAL		INFLIGHT (continued)	
AOA/FLAPS FAILURE	1	ELECTRICAL SYSTEM FAILURE	15
CADC/PITOT-STATIC MALFUNCTION	2	ENGINE FAILURE IN FLIGHT /	
STATIC INVERTER FAILURE	3	ENGINE FAILURE AT LOW ALTITUDE	16
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ENGINE FIRE DURING START	5	MALFUNCTION	18
TAKEOFF		FUEL LEAK	19
ABORT TAKEOFF	6	HYDRAULIC SYSTEMS FAILURE	20
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		LANDING WITH TIRE FAILURE	30
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CAUTION LIGHT	CONDITION	CORRECTIVE ACTION	
FIRE WARNING	Engine fire or overheat.	See FIRE WARNING procedures.	D T
AIR DATA COMPUTER	CADC outputs unreliable.	See CADC/PITOT-STATIC MALFUNCTION /2.	D T
AOA / FLAPS	AOA switching unit failure.	See AOA/FLAPS FAILURE /1.	D T
CANOPY	A) Canopy unlocked. B) Canopy lost or jettisoned.	A) Lock canopy. B) See LOSS OF CANOPY /21.	D
ENGINE ANTI-ICE ON	Anti-ice system operating.	Advisory.	
EXT TANKS EMPTY	External tanks empty.	Advisory.	
FLIGHT HYD UTILITY HYD	Hydraulic pressure in indicated system 1500 psi or less.	See HYDRAULIC SYSTEMS FAILURE /20.	D
L FUEL LOW R FUEL LOW	Usable fuel remaining in indicated system 400 lbs or less.	Check fuel balance.	
L FUEL PRESS R FUEL PRESS	Fuel pressure in indicated system 6.5 psi or less.	Check boost pumps ON, reduce RPM, descend below 25,000 feet. Monitor fuel gages. See FUEL LEAK /19 if a fuel leak is suspected.	D T
L GENERATOR R GENERATOR	Indicated generator not operating.	Turn ON or RESET generator. See ELECTRICAL SYSTEM FAILURE /15.	D T
OXYGEN	Oxygen remaining 0,5 liter or less, or pressure 40 psi or less.	Descend to a safe altitude, monitor supply pressure.	

BOLDFACE TEXT: critical steps performed immediately from memory.
D)amage, T)rigger: possible failure causes.



GENERAL

AOA / FLAPS FAILURE ^{DT}

1

Symptom: AOA/FLAPS caution.

Effect: Loss of AUTO flaps operation. Flaps remain in the position attained at the time of failure until another setting is selected.

1. Flap Thumb Switch — FXD or UP.
2. (Approach and landing) Flap Lever — FULL.

CADC / PITOT STATIC ^{DT}
MALFUNCTION

2

Symptoms:

AIR DATA COMPUTER caution — CADC failure.
Loss of airspeed gauge — pitot-static leakage.

Effects of CADC failure:

Loss of altimeter primary ELECT mode.
Loss of AUTO and FXD flap modes.
Loss of landing gear warning.

Effects of pitot-static malfunction:

Loss of airspeed gauge.
Loss of aux intake doors control.
Inadequate landing gear warnings.

1. Altimeter — Standby PNEU Mode.
2. Flap Thumb Switch — UP.
3. (Approach and landing) Flap Lever — FULL.
4. Aux Door circuit breakers — As Desired.

Failed pitot-static causes aux doors to open.
Pulling L ENG AUX DOOR and R ENG AUX DOOR circuit breakers closes doors and prevents their cycling that may lead to unexpected loss of thrust.

STATIC INVERTER FAILURE ^{DT}

3

Symptom: Failure of fuel and oxygen indicators to respond during GAGE TEST & QTY CHECK.

Effect: Engine ignition not available when aircraft is on battery power.

Starting an engine with failed static inverter requires an operating generator or connection of Ground Electric Power.

Avoid maneuvers that may lead to dual engine flameout, as airstart will not be possible.

GROUND OPERATIONS

EMERGENCY EXIT ON GROUND

4

1. Canopy — Open or jettison.
2. Throttles — OFF.

ENGINE FIRE DURING START ^{DT}

5

1. Throttles — OFF.
If engines fail to shut down:
2. Fuel Shutoff Switches — CLOSED.

TAKEOFF

ABORT TAKEOFF

6

1. **THROTTLES — IDLE.**
2. **CHUTE — DEPLOY.**
3. **WHEEL BRAKES — APPLY MAXIMUM.**
If arrestment need is evident:
4. **HOOK — DOWN.**

EMERGENCY JETTISON

7

1. EMER ALL JETT Button — Push.
If stores fail to jettison:
2. SELECT JETTISON Switch — ALL PYLONS.
3. SELECT JETTISON Button — Push.

ENGINE FAILURE/FIRE
WARNING DURING TAKEOFF ^{DT}

8

If takeoff is refused:

1. Abort /6.
If engine fire is suspected, place the throttle of the affected engine to OFF.
If the fire is confirmed, perform Emergency Exit On Ground /4 procedure after stopping.
- If takeoff is continued:
1. **THROTTLES — MAX.**
2. **STORES — JETTISON (IF NECESSARY) /7.**
3. At safe ejection altitude (2000 ft AGL) — Perform ENGINE FAILURE IN FLIGHT /16 and/or FIRE WARNING procedures.

LANDING GEAR ^D

RETRACTION FAILURE

9

Retraction of the main gear requires at least 1500 psi in the utility hydraulic system.

If the landing gear warning light remains on after lever has been moved to LG UP:

1. **Airspeed — Maintain below 260 kts.**
2. **Gear Alternate Release Reset Control — Cycle.**
3. **Nose Strut — Retracted (switch in aft pos).**
4. **Gear Lever — LG DOWN, Then LG UP.**
5. **Throttles — cycle to MIL.**
6. **If light remains on — Lower gear.**

If light goes out, proceed with flight.

Note that a damaged gear may fail to retract, but its indicator light may still go out.

SINGLE ENGINE TAKEOFF

10

If an engine fails and takeoff is continued:

- Use MAX thrust until reaching the safe ejection altitude (2000 ft AGL).
- Jettison stores if acceleration is inadequate.
- Use remaining runway to increase airspeed as much as possible.
- Immediately after liftoff, focus on accelerating, not on gaining altitude, until confident that the aircraft can maintain a minimum climb rate of 300 fpm.
- Raise landing gear after a steady rate of climb is attained, but no later than 210 kts.
- Recommended single-engine climb speeds:

GEAR DOWN, FLAPS AUTO - 210 kts
GEAR UP, FLAPS AUTO - 230 kts
GEAR UP, FLAPS UP - 290 kts

TIRE FAILURE ON TAKEOFF ^D

11

If takeoff is refused:

1. **Abort /6.**

If takeoff is continued:

1. **GEAR — DO NOT RETRACT.**
2. **See LANDING WITH TIRE FAILURE /30.**

AIRSTART / MAXIMUM GLIDE

12

- | If both engines flame out with inoperative static inverter /3, restart is not possible.
- | Frozen (0% RPM) engine cannot be restarted.

Procedure applies to affected engine. If both engines flame out, left engine start should be attempted first.

1. **Throttle — OFF.**
2. **Altitude — Below 30,000 feet.**
3. **Airspeed — 250 kts (optimal).**
4. **Boost Pump Switches — LEFT and RIGHT.**
5. **Battery Switch — Check BATT.**
6. **Start Button — PUSH.**
7. **Throttle — IDLE.**

If restart does not occur within 50 seconds, increase engine windmill speed by increasing airspeed and/or decreasing altitude before attempting another start.

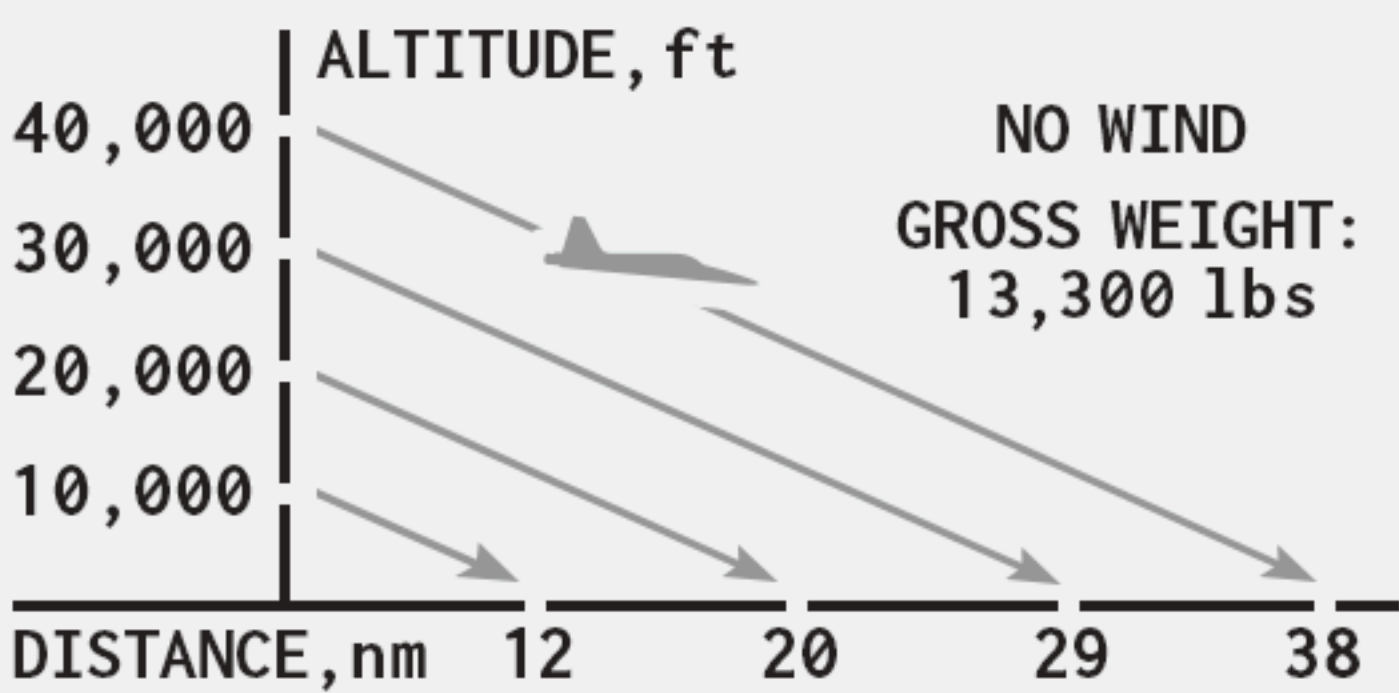
■ ALTERNATE AIRSTART

1. **Altitude — Below 30,000 feet.**
2. **Airspeed — 250 kts (optimal).**
3. **Battery Switch — Check BATT.**
4. **Throttle — Advance to MAX.**
This activates ignition. If the throttle is already at MAX, cycle it to MIL and return to MAX.

■ MAXIMUM GLIDE (BOTH ENGINES WINDMILLING)

1. **Flaps — Up.**
2. **Stores — Jettisoned.**
3. **Airspeed — 250 kts (optimal for airstart).**

Approximate glide distance for each 1000 ft AGL, with or without pylons: 1 nm.



CONTROLLABILITY CHECK

13

If the aircraft's normal performance has been degraded by damage, failures, or other factors:

1. **Altitude** — 15,000 feet AGL (if practical).
2. **Landing Configuration** — Establish.
Minimize flap movement if flap damage is known or suspected.
3. **Airspeed** — Reduce to determine the acceptable approach and landing characteristics.

■ APPROACH AND LANDING

1. **Gear and Flaps** — As determined.
Do not change landing configuration.
2. **Landing Approach** — Straight In.
Plan to fly a power-on, straight in approach requiring minimum flare.
3. **Landing Speed** — No slower than determined airspeed.
4. **Touchdown** — At or above determined airspeed.

EJECTION

14

- Ejection is preferable to landing on an unprepared surface. Do not attempt landing with both engines flamed-out.
- Eject before the start of any sink rate.
- If possible, eject above 2000 ft AGL flying straight and level at low airspeed.
- When below 2000 ft AGL, zoom to trade speed for altitude, and eject before climb rate reaches zero.
- Under uncontrolled conditions (spins, dives) eject at least 10,000 ft AGL whenever possible.

■ BEFORE EJECTION

1. IFF to EMER. UHF to GUARD. Transmit mayday, giving position and intentions.
2. Turn aircraft toward uninhabited area.
3. Wings level. Slow down as much as practical. Attempt a positive rate-of-climb.

ELECTRICAL SYSTEM FAILURE ^{D T}

15

■ ALTERNATE AIRSTART

Symptom: L or R GENERATOR caution.

Effect: Loss of control over the corresponding aux intake door (door remains closed).

The remaining generator picks up the load.

| If one engine is frozen (0% RPM), the remaining engine requires RPM above idle (approx. 68%) to keep generator on line.

■ AC / COMPLETE ELECTRICAL FAILURE

Symptom: L and R GENERATOR cautions.

Effect: The following are inoperative with complete electrical failure:

- FLIGHT AND ENGINE INSTRUMENTS
 - EXCEPTION:**
 - TACHOMETERS
 - AIRSPEED INDICATOR
 - ALTIMETER IN PNEU MODE
 - STANDBY ATTITUDE INDICATOR
 - VERTICAL VELOCITY INDICATOR
 - ACCELEROMETER
 - ARRESTING HOOK EXTENSION
 - AUX INTAKE DOORS
 - FLAPS
 - FUEL BOOST PUMPS
 - OPTICAL SIGHT
 - PITCH AND AILERON TRIM
 - RADAR
 - RADAR WARNING RECEIVER
 - STABILITY AUGMENTER SYSTEM
 - TACAN, ADF
1. **Throttle(s)** — Retard (if above 25,000 ft).
 2. **Altitude** — 25,000 ft or below.
 3. **Battery Switch** — Check BATT.
 4. **Generator Switch(es)** — Reset, then L GEN/ R GEN.
 5. **Crossfeed** — As Req.
 6. **Altimeter** — Standby PNEU Mode.
 7. **Land as soon as possible.**
- Operate only systems necessary for flight and landing.

INFLIGHT (continued)

ENGINE FAILURE IN FLIGHT /
ENGINE FAILURE AT LOW ALTITUDE

16

Effects of RIGHT engine failure: Loss of right generator and flight control hydraulic pump.

Left generator picks up the load. Utility system provides hydraulic power for the flight controls.

Effects of LEFT engine failure: Loss of left generator and utility hydraulic pump. Any flight control movement rapidly depletes hydraulic pressure in the utility system.

Right generator picks up the load. Some hydraulic pressure is regained from windmilling RPM, but at a rate insufficient for reliable operation of the following:

- LANDING GEAR NORMAL EXTENSION
- NOSEWHEEL STEERING
- SPEED BRAKE
- STABILITY AUGMENTER SYSTEM
- WHEEL BRAKES

■ ENGINE FAILURE IN FLIGHT

1. **Throttle (good engine) — As Req.**
2. **Stores — Jettison (if necessary).**
3. **Gear / Speed Brake — Up / In.**
4. **Throttle (failed engine) — OFF.**
5. **Fuel Balancing — As Req.**
Autobalance should be used, if available.
With fuel less than 400 lbs in each system:
6. **Fuel Boost Pumps — LEFT and RIGHT.**
7. **Crossfeed Switch — CROSSFEED.**

■ ENGINE FAILURE AT LOW ALTITUDE

1. **THROTTLES — MAX.**
2. **Engine Instruments — Monitor.**
If both engines fail at low altitude and with sufficient airspeed:
 - Zoom to trade speed for altitude.
 - Try to airstart immediately **/12.**
 - Ejection should be done while in a positive climb.
 - If continued airstarts are to be attempted, lower the nose before airspeed drops below 250 kts.

FIRE WARNING IN FLIGHT ^{DT}
(AFFECTED ENGINE)

17

1. **THROTTLE — IDLE.**
2. **THROTTLE — OFF IF FIRE WARNING LIGHT REMAINS ON.**
Close fuel shutoff switch if engine fails to shut down or if fire warning light remains on.
3. **IF FIRE IS CONFIRMED — EJECT **/14.****

FUEL AUTOBALANCE
SYSTEM MALFUNCTION ^{DT}

18

■ AUTO BALANCE SWITCH STUCK
IN CENTER POSITION

Autobalance system is considered inoperative.

1. **Use manual balancing (as required).**

■ AUTO BALANCE SWITCH STUCK
IN LEFT LOW OR RIGHT LOW POSITION

With this failure, the autobalance system continuously draws fuel from the selected fuel system until it is empty. After that, the fuel from the opposite system is drawn by gravity flow.

This procedure puts both fuel systems in gravity feed mode after their fuel quantity has been evened by autobalance.

1. **Fuel Quantity — Check often.**
Once both systems are balanced:
2. **Altitude — 25,000 ft or below.**
3. **Crossfeed Switch — CROSSFEED.**
4. **Fuel Boost Pump on the side opposite to where the Auto Balance switch is stuck — OFF.**

L FUEL PRESS and R FUEL PRESS caution lights indicate gravity feed from both fuel systems.

5. **Land as soon as practical.**
Fly with reduced power at lowest practical altitude to assure continued stable engine operation with boost pumps inoperative.
If required, perform external fuel transfer.

FUEL LEAK ^D

19

Damage to the internal fuel tanks can result in a rapid and evident decrease in fuel quantity in one or both fuel systems.

Once occurred, fuel leaks cannot be stopped or reduced. Depending on the severity of the damage, the system may lose the remaining fuel and cause engine flameout in less than one minute.

Because fuel leaks don't have dedicated warnings, it is important to check fuel quantity each time:

- Note
- After egress/fencing out;
 - When the aircraft sustains damage;
 - When the FUEL PRESS caution illuminates.

■ IMMEDIATELY UPON DETECTION OF RAPID FUEL LOSS

1. **THROTTLES — MAX.**
2. **CLIMB ABOVE SAFE EJECTION ALTITUDE.**
3. **DETERMINE THE AFFECTED FUEL SYSTEM.**

■ BOTH SYSTEMS AFFECTED

1. Trade remaining fuel for altitude and time.
2. Prepare for controlled ejection after both engines flame out.

If speed and altitude permit, an airstart using fuel from the external tanks may be attempted as a last resort to save the aircraft.

■ ONE SYSTEM AFFECTED

1. Boost Pump on the affected side — OFF.
2. Crossfeed Switch — CROSSFEED.
This might prevent engine flameout when the affected system runs out of fuel.
3. Throttles — As required to maintain at least 250 kts.
4. Climb to a safe altitude, but below 25,000 ft.
5. Land as soon as practical.
If needed, an airstart and/or fuel transfer from the external tanks may be attempted.

HYDRAULIC SYSTEMS FAILURE ^D

20

Hydraulic system failure is indicated by a total and irreversible loss of hydraulic pressure in the affected system.

Note

It should not be confused with pressure fluctuations caused by the loss of hydraulic pump following engine failure.

■ DUAL SYSTEM FAILURE

Symptoms: FLIGHT HYD and UTILITY HYD cautions. Loss of pressure in both systems.

1. If flight control becomes impossible — Eject.

■ FLIGHT CONTROL SYSTEM FAILURE

Symptoms: FLIGHT HYD caution. Loss of pressure in the flight control system.

The utility system has enough capacity to provide hydraulic power for the flight controls.

1. Land as soon as practical.

■ UTILITY SYSTEM FAILURE

Symptoms: UTILITY HYD caution. Loss of pressure in the utility system.

Effect: The following are inoperative with utility system failure:

- GUN GAS DEFLECTOR AND BAY PURGE DOORS
- LANDING GEAR NORMAL EXTENSION
- NOSEWHEEL STEERING
- SPEED BRAKE
- STABILITY AUGMENTER SYSTEM
- WHEEL BRAKES

1. Pitch and Yaw Damper Switches — OFF.
2. Land as soon as practical.

LOSS OF CANOPY ^D

21

If the canopy is lost or jettisoned:

1. Reduce airspeed to 300 kts or less.
2. Oxygen Supply Lever — ON.

If the oxygen supply is not available, descend to altitude not requiring oxygen.

PITCH DAMPER FAILURE ^T

22

Symptoms: Applying aft or forward stick results in excessive oscillations in pitch that take longer than usual to settle.

- 1. Pitch Damper Switch — Verify in PITCH.
- 2. (When carrying external tanks) Airspeed — Reduce below 0,75 M.

PITCH TRIM FAILURE ^T

23

- 1. Airspeed — Adjust (as necessary to minimize stick forces).
- 2. Controllability check — Perform /13. Approach and landing — perform as instructed.

SPIN & POST-STALL GYRATION RECOVERY

24

■ ERECT PSG RECOVERY

When uncontrolled aircraft oscillations at angles of attack above stall are encountered:

- 1. Relax aft stick pressure to reduce AOA. If this does not immediately resolve PSG:
- 2. **STICK — FORWARD AS REQUIRED.** Delayed application may result in spin entry.
- 3. Ailerons and Rudder — Neutral. Recovery is indicated when the aircraft responds to forward stick and airspeed exceeds 130 kts.
- 4. Relax forward stick after recovery. Smoothly apply aft stick to regain positive G flight.

Continued application of forward stick may result in inverted PSG or inverted spin entry. Failure to apply aft stick may result in the inverted pitch hangup (IPH).

PSG recovery may require 4000 feet of altitude, and approx. 6000 feet for dive pullout.

■ ERECT SPIN RECOVERY

Immediately upon recognition of spin direction:

- 1. **STICK — FULL FORWARD.**
- 2. **AILERON — FULL IN DIRECTION OF SPIN** (bypass the Aileron Limiter).
- 3. **RUDDER — FULL OPPOSITE.**

- 4. Flaps — AUTO. Spin recovery begins with gradual increase of nose-low attitude, and ends in a strong pitchover when airspeed increases through 130 kts.
- 5. Neutralize controls after recovery. Smoothly apply aft stick to regain positive G flight. Altitude loss per turn is approx. 1700–2500 feet. Spin recovery requires a minimum of two turns and 4000 feet of altitude. Dive pullout requires approx. 6000 feet.

■ INVERTED PSG / INVERTED SPIN RECOVERY

- 1. **FLAPS — UP.**
- 2. **STICK — AFT AS REQUIRED.**
- 3. **AILERONS AND RUDDER — NEUTRAL.** Recovery is indicated by the decrease of negative G and the onset of positive G.
- 4. Relax aft stick after recovery. If considerable aft stick is maintained after recovery, entry into an erect PSG or spin is possible. Altitude loss during inverted PSG or spin may exceed 6000 feet (not including dive pullout).

Note | A windmilling engine has a very high probability of freezing (RPM drops below 10%) during stalls and spins, rendering restart impossible.

LANDING

ARRESTMENT (APPROACH END)

25

None of the airfields in the operational area are equipped with functional arresting gear. This procedure is intended for practicing.

- 1. Landing configuration — Establish.
- 2. Hook — Down.
- 3. AOA — On Speed.
- 4. Touchdown — In center of runway, minimum 500 feet before cable.
- 5. Chute — Deploy. Discontinue braking before crossing cable.
- 6. (After engagement) Throttles — As required to control rollback.

LANDING (continued)

DRAG CHUTE FAILURE

26

If the drag chute fails to deploy on landing, and the decision is made to go around:

1. Drag Chute — Jettison.
2. Go Around /See CHECKLIST-II.

SINGLE-ENGINE APPROACH & LANDING

27

■ SINGLE-ENGINE APPROACH

- Delay lowering landing gear until just before glide path. MAX thrust should be used on single-engine approaches if necessary.
- Expend or jettison stores before entering the landing pattern, if necessary.
- Smoothly apply the rudder to counter yaw. Avoid abrupt control inputs, as the aircraft in this configuration is very susceptible to spin entry.

■ SINGLE-ENGINE LANDING

1. Flap Thumb Switch — AUTO.
2. Gear — Down.

With a failed left engine, windmilling RPM may be sufficient for normal gear extension that requires at least 2700 psi in the utility system. However, if gear does not extend, use the Alternate Release Handle. Wheel brakes and nosewheel steering are not available.

3. Airspeed — Increase 10 kts above normal until landing assured.
4. AOA Indicator — 14.0 units on final approach.
5. Drag Chute — As required.

■ SINGLE-ENGINE MISSED APPROACH

- Use MAX thrust.
- Raise landing gear after a steady rate of climb is attained.
- Keep flaps in AUTO.
- Climb at 260 kts with landing gear up or 210 kts with landing gear down.

LANDING GEAR ALTERNATE EXTENSION

28

1. Airspeed — 260 kts or less.
2. Gear Lever — LG DOWN.
3. Gear Alternate Release Handle — Pull.
Alternate extension takes approx. 15 seconds.
4. Gear Indicators — Check three green.

If the main gear fails to extend fully, rocking wings or yawing the aircraft aid the extension.

If repeated extension attempts are unsuccessful, see **Landing Gear Extension Failure /29**.

Normal extension of the main gear requires at least 2700 psi in the utility system.

Note To resume normal operation of the Gear Lever after using Alternate Release Handle, cycle the Alternate Release Reset Control.

LANDING GEAR EXTENSION FAILURE

29

Use the following table to determine whether a landing with a particular gear condition is feasible, or ejection is the best course of action:

RECOMMENDED ACTION DEPENDING ON THE ACTUAL (NOT INDICATED) GEAR POSITION					
NOSE	UP	UP	UP	DOWN	DOWN
MAIN	BOTH DOWN	BOTH UP	ONE DOWN	BOTH UP	ONE DOWN
	LAND	EJECT	EJECT		
UNLESS LOADOUT ALLOWS BELLY LANDING					

Fly a normal landing pattern.
If possible, reduce gross weight by expending excess fuel and jettison armament.

If belly landing is to be attempted, retain empty fuel tanks on wing and/or centerline pylons to cushion shock of nose slam-down. If not possible, attain a clean no pylon configuration.

Note Gear is considered unsafe when damaged or stuck in a partially extended position. Unsafe gear may collapse at touchdown.

(Continued on the next page)

LANDING (continued)

LANDING GEAR
EXTENSION FAILURE (CONTINUED)

■ LANDING WITH NOSE GEAR UP OR UNSAFE

- 1. Throttles — IDLE at touchdown.
- 2. Nose — Gently lower to runway.
If nose gear is up, position throttles OFF when nose contacts runway.
- 3. Drag Chute — Deploy.
Do not use brakes if a safe stop can be made without them when the nose gear is unsafe.
- 4. Battery Switch — OFF.

■ BELLY LANDING

Without empty fuel tanks:

- 1. Eject.
With empty fuel tanks:
- 1. Gear — Up.
- 2. Throttles — OFF at touchdown.
- 3. Drag Chute — Deploy when on runway.
- 4. Battery Switch — OFF.
With no pylons installed:
- 1. Gear — Up.
- 2. Speed Brake — Open.
- 3. Flap Thumb Switch — AUTO.
- 4. Airspeed — Increase touchdown speed 10 kts.
Fly a power on approach requiring minimum flare.
- 5. Throttles — OFF at touchdown.
- 6. Drag Chute — Deploy when on runway.
- 7. Battery Switch — OFF.

■ LANDING WITH ONE OR BOTH MAIN GEAR UP

If both main gear cannot be extended
with nose gear up or down
and all gear cannot be retracted:

- 1. Eject.
If landing must be attempted:
- 1. Throttles — IDLE at touchdown.
With one main gear extended, touch down in center of runway. Use aileron to hold wings level.
- 2. Drag Chute — Deploy.
- 3. Throttles — OFF.
- 4. Battery Switch — OFF.

LANDING WITH TIRE FAILURE ^D

30

- Fly a normal traffic pattern.
- Land beyond the approach-end arresting cables.

■ NOSE GEAR

- Expend excess fuel, fire out ammunition, and jettison centerline store to obtain a more favorable aft CG position.
- After touchdown, hold nosewheel off runway as long as possible. When nosewheel touches down, deploy drag chute.
- Nosewheel steering angle must not exceed 20 degrees left or right.

■ MAIN GEAR

- Braking efficiency on the ground is reduced.
- Expend excess fuel and jettison stores, but retain empty pylon fuel tanks.
- Land on the side of the runway away from the failed tire. When it is unknown which tire is failed, touch down in the center of the runway.
- After touchdown, deploy drag chute.

NO-FLAP LANDING &
FLAP ASYMMETRY

31

■ NO-FLAP LANDING

When landing with the wing flaps retracted, use the normal landing procedure modified as follows:

- 1. Pattern — Fly wider than normal.
- 2. Final turn, final approach, and touchdown airspeeds — Increase by 10 kts.
- 3. AOA Indicator — 16.4 units on final approach.
Landing distance increases approximately 15%.

■ FLAP ASYMMETRY

When landing with a flap asymmetry, including one flap damaged or lost:

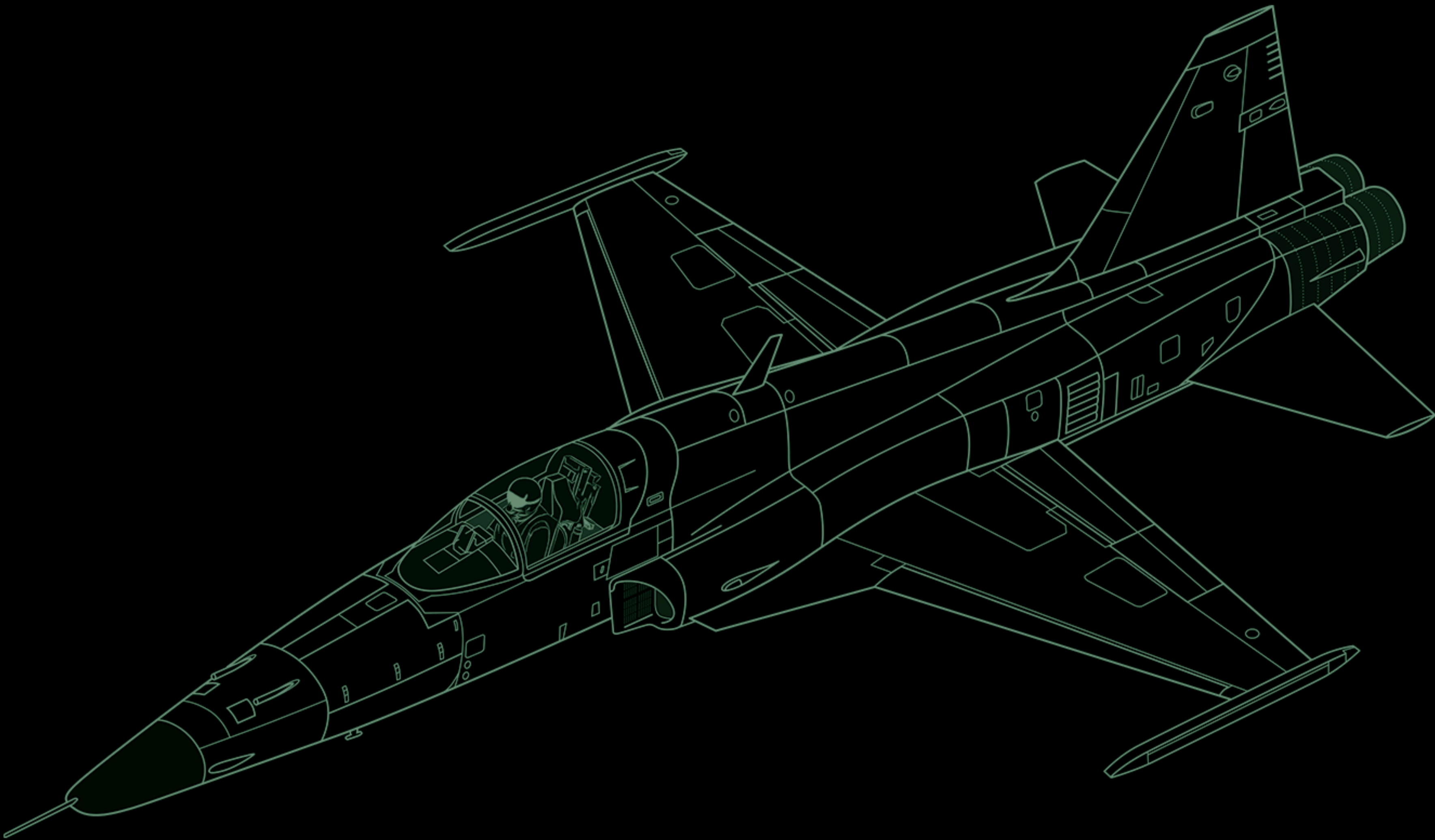
- 1. Controllability check — Perform /13.
Use UP or FULL flaps. Do not change flap position during approach and go-around.
- 2. Follow the No-Flap Landing procedure.



(SIMPLIFIED)

KNEEBOARD
SUITE

F-5E-3



Checklist-I

Checklist-II and Combat Checks

RWR Threats

Ordnance and Fuel Balancing

Approach and Landing Patterns

Appendix-I

Appendix-II

Cockpit and Quickstart

Emergencies and Cautions

Cockpit Diagram (Large)

2

3

4

5

6

7

8

9

10-18

19

PREFLIGHT CHECK*

1. Ground Air Supply — Connect.
2. Pins and Covers — Check Removed.
3. Flap Lever — THUMB SW.
4. Speed Brake Switch — Neutral (OFF).
5. Flap Thumb Switch — UP.
6. Standby Attitude Indicator — Uncage (ccw).
7. Flight Instruments, Clock — Check and Set.
8. UHF, TACAN, NAV MODE — As Req.
9. Fuel Boost Pumps — LEFT and RIGHT.
10. Battery Switch — BATT.
11. Generators — L GEN and R GEN (rmb).
12. Fuel & Oxy Switch — GAGE TEST, QTY CHECK.
No indicator response indicates static inverter failure.
13. IFF / Master Switch — As Req / STBY.
14. Int and Ext Lights, Beacon — As Req.

STARTING ENGINES LEFT ENGINE

1. Ground Air Supply — Apply.
2. At 10% RPM, Start Button — PUSH.
3. Throttle — Advance to IDLE.
4. Engine RPM — Stabilized at 49–52%.
5. Hydraulic Pressure — Check 2800–3200 psi.
6. Generator Caution Light — Out.

GROUND AIR START RIGHT ENGINE

Omit this procedure if crossbleed start is to be used.

1. Same as for Left Engine.
2. Ground Air, Ground Power — Disconnect.

CROSSBLEED START RIGHT ENGINE

Recommended be used only in isolated areas.

1. Ground Air, Ground Power — Disconnect.
2. Wheel Chocks — Place.
3. Left Engine RPM — Increase to 95%.
4. Right Engine Start Button — PUSH.
5. At 10% RPM, Right Throttle — IDLE.
6. Left Throttle — Retard to Idle after Right Engine is stabilized at 49–52% RPM.
7. Generator Caution Lights — Out.

BEFORE TAXI

1. Speed Brake — In.
2. Flap Thumb Switch — AUTO (verify flaps at full).
3. Damper Switches — YAW and PITCH.
4. Pitch Trim — Set (refer to the table below).
5. Altimeter — To QNH (or to match field elevation).
6. Primary and Standby Attitude Indicators — Set 3 Degrees Nose Low.
7. Apply brakes; Wheel Chocks — Remove.

BEFORE TAKEOFF

For dehiked strut takeoff, expect up to 20% increase in V_R and up to 45% increase in takeoff roll.

1. Nose Strut Switch — EXTEND.
2. Optical Sight Mode Selector — As Req.
3. Radar Warning Receiver — As Req.
4. Radar Mode Selector — As Req.
5. Pitot Heat, Engine Anti-Ice — As Req.
6. HSI/ IFF — To runway course / As Req.
7. Canopy — Closed.
8. Caution and Warning Lights — Out.
9. Ext Lights, Beacon — As Req.

TAKEOFF

Do not engage nosewheel steering above 65 knots.

1. Apply brakes. Set MIL power RPM.
2. Release brakes. MAX thrust — As Req.
3. Aft stick — At 10 knots below liftoff speed.

WEIGHT ×1000 lbs	TAKEOFF DATA		LIFTOFF SPEED	PITCH TRIM	
	LOADOUT				
≤ 15	CLEAN, NO AMMO	143-145	6		
15,5-16	G M	153-155	7		
17-18	G M 1×F	164-168	7		
19	G M 3×F ¹⁵⁰	166-168	7		
19-21	G M 1×F B R	168-175	8		
22	G M 3×F	178-180	8		
≥ 23	G M B R	185-190	9		

* Kneeboard adjustments require cold and dark aircraft with battery off. Ground crew requires either an open canopy, battery power, or ground power.

If gun ammo is removed from loadout, decrease the liftoff speed by 10 kts.

G)un ammo
M)issiles
F)uel tanks
B)ombs
R)ockets

AFTER TAKEOFF. CLIMB.

- 1. Gear — Up.
- 2. Flap Thumb Switch — As Req.
- 3. External Fuel/Autobalance — As Req.
- 4. Oxygen, Cockpit Pressurization — Check.
- 5. Altimeter — As Req.

DESCENT

- 1. Armament Safety Check — Complete.
- 2. Canopy Defog, Engine Anti-Ice, and Pitot Heat — As Req.
- 3. Oxygen — Check.
- 4. Altimeter — Check and Set.
- 5. Ext and Landing Lights — As Req.

BEFORE LANDING

- 1. Altimeter — Check and Set.
- 2. Manual Crossfeed — Discontinue.
- 3. Hydraulic Pressure — 2800–3200 psi.
- 4. Flap Thumb Switch — AUTO.
- 5. Gear — Down (below 260 kts).
- 6. AOA — On Speed.

If AOA is inop, maintain 145 kts + weight correction.
Add 5 kts for full guns ammo. Add 1 kts for each 200 lbs of fuel above 1000 lbs remaining.

Drag chute max deployment speed: 180 kts.

GO-AROUND

- 1. *Throttles — MIL (MAX if necessary).*
- 2. *Speed Brake — In.*
- 3. *Confirm positive rate of climb; Gear — Up.*
- 4. *Flaps — As Req.*

ARMAMENT SAFETY CHECK

- 0. Fuel Quantity — Check.
 - 1. Guns, Msl & Camr Switch — OFF.
 - 2. Bombs Arm Switch — SAFE.
 - 3. Armament Select Switches — OFF.
 - 4. External Stores Selector — SAFE.
 - 5. Sight Mode Selector — OFF.
 - 6. Missile Volume — Off (ccw).

COMPLETE AS SOON AS PRACTICABLE
AFTER WEAPONS EMPLOYMENT

AFTER LANDING

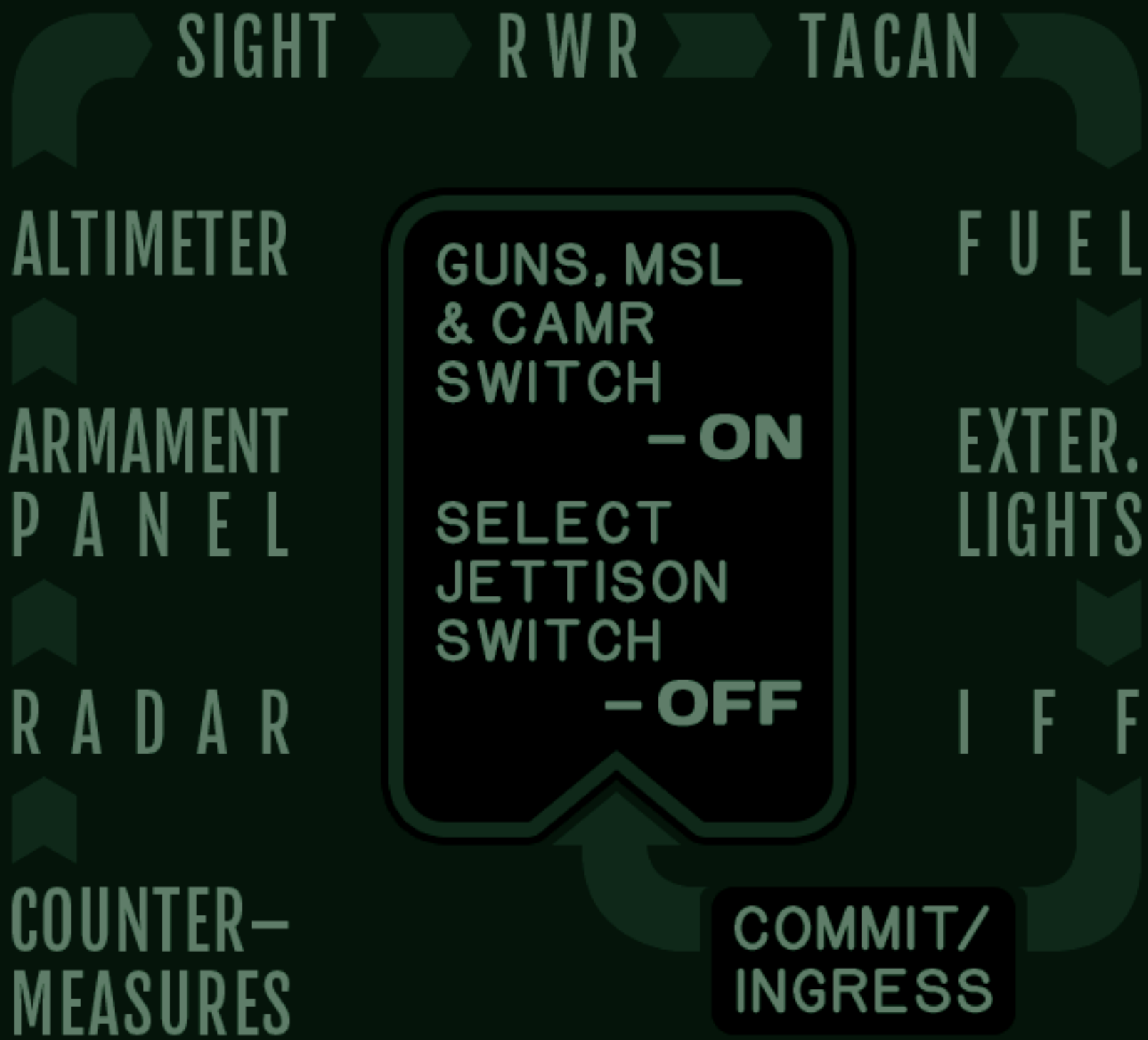
- 1. Drag Chute — Jettison (if deployed).
Prior to opening canopy:
- 2. Check Cabin Pressure Altimeter;
Cockpit Pressurization Switch — RAM DUMP.
- 3. Flap Thumb Switch — UP.
- 4. Speed Brake — Out.
- 5. Optical Sight, RWR, and Radar — OFF.
- 6. Pitot Heat and Engine Anti-Ice — OFF.
- 7. Ext and Landing Lights, Beacon — As Req.
- 8. IFF — OFF.
- 9. Pitch Trim — Reset (6 to 7 increments).

ENGINE SHUTDOWN

- 1. Canopy — Open before shutdown.
- 2. Cockpit Pressurization Switch — NORMAL.
- 3. Apply brakes; Wheel Chocks — Place.
- 4. All Unguarded Switches (except Battery, Generators, and Fuel Boost Pumps) — OFF.
- 5. Seat — Full Up.
- 6. Throttles — OFF.
- 7. Standby Attitude Indicator — Cage (pull + cw).
- 8. Battery Switch — OFF.

FLAP THUMB SWITCH

- UP: Max range. Max endurance without stores.
- FIXED: Max endurance with stores.
- AUTO: Max turn performance. For all phases of maneuvering flight. Not for enroute cruise.



COMBAT CHECKS

4 RWR Threats				F-5E-3 DimOn							
14	F-14A/GR/B	123 88	21	MiG-21Bis	12 09	37	AJS-37 Viggen	2 06	JF	JF-17	23 46
15	F-15C	123 49	23	MiG-23MLD	12 14	50	A-50 / KJ-2000	AWACS	M2	M-2000C	12 26
	F-15E	23 49	25	MiG-25PD	12 25	64	AH-64D BLK.II	3 05		Mirage 2000-5	123 24
	F-15E S4+	123 49	29	MiG-29A/G	12 38		(with or w/o MMA)		U	A-6E	2 04
16	F-16A	123 38		MiG-29S	123 33	E2	E-2D	AWACS		AH-64D (old)	3 05
	F-16A MLU	23 38		Su-27/33	12 32	E3	E-3A	AWACS		C-101CC	2 04
	F-16C/CM	23 38		J-11A	123 32	F1	Mirage F1	12 22		H-6J	BOMBER
18	F/A-18A	12 27	30	Su-30	123 37	F2	Tornado GR4	2 04		Tornado IDS	2 04
	F/A-18C	123 42	31	MiG-31	12 37	F4	F-4E-45MC	12 27		Tu-95MS	BOMBER
19	MiG-19P	2 03	34	Su-34	123 35	F5	F-5E/E-3/FC	2 04		SH-60B	HELO

SURFACE					NAVAL			
S	Nebo 1L13 / Nebo 55G6 / AN/FPS-117		EWR		15	Project 22160 Tor-M2 CORVETTE		C 10/39
	Flat Face B	[2-3-5]	SR		40	LHA-1 Tarawa CV		F 08/13
BB	Big Bird	[10]	long-range	SR	49	Oliver Hazard Perry FRIGATE		C 26/++
CS	Clam Shell	[10]	low-altitude	SR	AE	Arleigh Burke DESTROYER		C 80/++
DE	Dog Ear Sborka	[13-15-19]	MCC-SR			Ticonderoga CRUISER		C 37/++
FC	Fire Can SON-9		AAA FDR		FR	CV-59 Forrestal CV		C 10/24
HQ	HQ-7B	[7]	SR		HN	Type 052C DESTROYER		C 90/++
SD	Snow Drift	[11]	SR			Pyotr Velikiy CRUISER		C 86/++
TS	Tin Shield	[5-10]	SR		HP	Grisha (2025) CORVETTE		C 07/17
U	Freya / Würzburg		WWII Assets Pack	EWR		Grisha (old) TO BE REMOVED		C 08/15
	Silkworm		anti-ship	SR	IV	HMS Invincible CV		C 47/++
	Tall King	[5]	SR		LC	La Combattante FAST ATTACK CRAFT		E 04/10
					MR	Type 052B DESTROYER		C 31/++
2	C 27/++ SA-2 Guideline S-75		Fan Song	TR		Type 054A FRIGATE		C 33/++
			Amazonka	RF	PS	Type 071 Amphib.dock TRANSPORT		E 06/10
3	C 13/++ SA-3 Goa S-125		Low Blow	TR		Molniya CORVETTE		E 03/09
5	C 160/+ SA-5 Gammon S-200		Square Pair	TR	SC	LS Ropucha BDK-775 TRANSPORT		E 06/09
6	C 18/27 SA-6 Gainful Kub		Straight Flush	STR	SS	CVN-71 Roosevelt CV		C 10/24 F 08/13
7	E 10/18 HQ-7B		TELAR			CVN-72 Lincoln CV		
8	C 07/17 SA-8 Gecko Osa		TELAR			CVN-73 Washington CV		
10	C 48/++ SA-10 Grumble S-300PS		Flap Lid A & B	TR		CVN-74 Stennis CV		
11	C 29/++ SA-11 Gadfly Buk		Fire Dome	TELAR		CVN-75 Truman CV		
13	F 03/12 SA-13 Gopher Strela-10M		TELAR		SW	Kuznetsov 2017 CV		C 08/20
15	C 09/20 SA-15 Gauntlet Tor		TELAR			Kuznetsov (old) CV		C 08/20
	C 10/39 SA-15 Gauntlet M2 Tor-M2		TELAR		T2	Moskva CRUISER		C 48/++
19	E 05/12 SA-19 Grison Tunguska		SPAAW			Project 22160 CORVETTE		E 04/13
22	C 11/33 SA-22 Greyhound Pantsir-S1		SPAAW		TP	Condell-class FRIGATE		E 02/07
A	E 03/10 Gepard		SPAAA			CNS Almirante Condell / CNS Almirante Lynch		
	E 02/06 Vulcan		SPAAA			Leander-class FRIGATE		E 06/11
	E 02/07 ZSU-23-4 Shilka		SPAAA			HMS Achilles / HMS Andromeda / HMS Ariadne		
CR	E 02/05 LPWS Centurion Phalanx		C-RAM			Neustrashimy FRIGATE		C 08/20
HK	C 14/++ Hawk		CWAR / SR / TR			Rezky FRIGATE		C 08/15
IT	F 22/++ IRIS-T SLM		STR		U	ARA Veinticinco de Mayo CV		E 02/10
NS	C 13/35 NASAMS AIM-120B		SR			Essex Class 1944 CV		E 02/07
	C 13/40 NASAMS AIM-120C		SR			Type 021-1 MISSILE BOAT		E 02/12
P	C 60/++ Patriot		STR					
RO	C 05/20 Roland		EWR / ADS					
RP	E 05/10 Rapier with Blindfire Radar		TR					
RT	E 05/10 Rapier with or without Tracker		LN					
U	E 10/20 HQ-7 'Player' AI- or user-controlled		TELAR					

1 — SARH, 2 — IR, 3 — ARH — SUPPORTED A-A MISSILES, FOLLOWED BY THE MAX LAUNCH RANGE AT 40.000 ft, nm.
18/27 — SAM'S MAX FIRING RANGE (nm) AND ALTITUDE (×1000 ft ABOVE GROUND LEVEL; ++ WHEN EXCEEDS 50.000ft).
CM TO USE: C — CHAFF, F — FLARES, E — EVADE. [THE SAM] TO WHICH THIS RADAR IS USUALLY LINKED.
ALL RANGES WERE MEASURED USING THE F/A-18C. F-5's RWR DOES NOT DISTINGUISH AIRBORNE AND SURFACE EMITTERS.

○ TRACKING (FLASHING — GUIDING THE MISSILE) ◇ PRIMARY THREAT M MISSILE



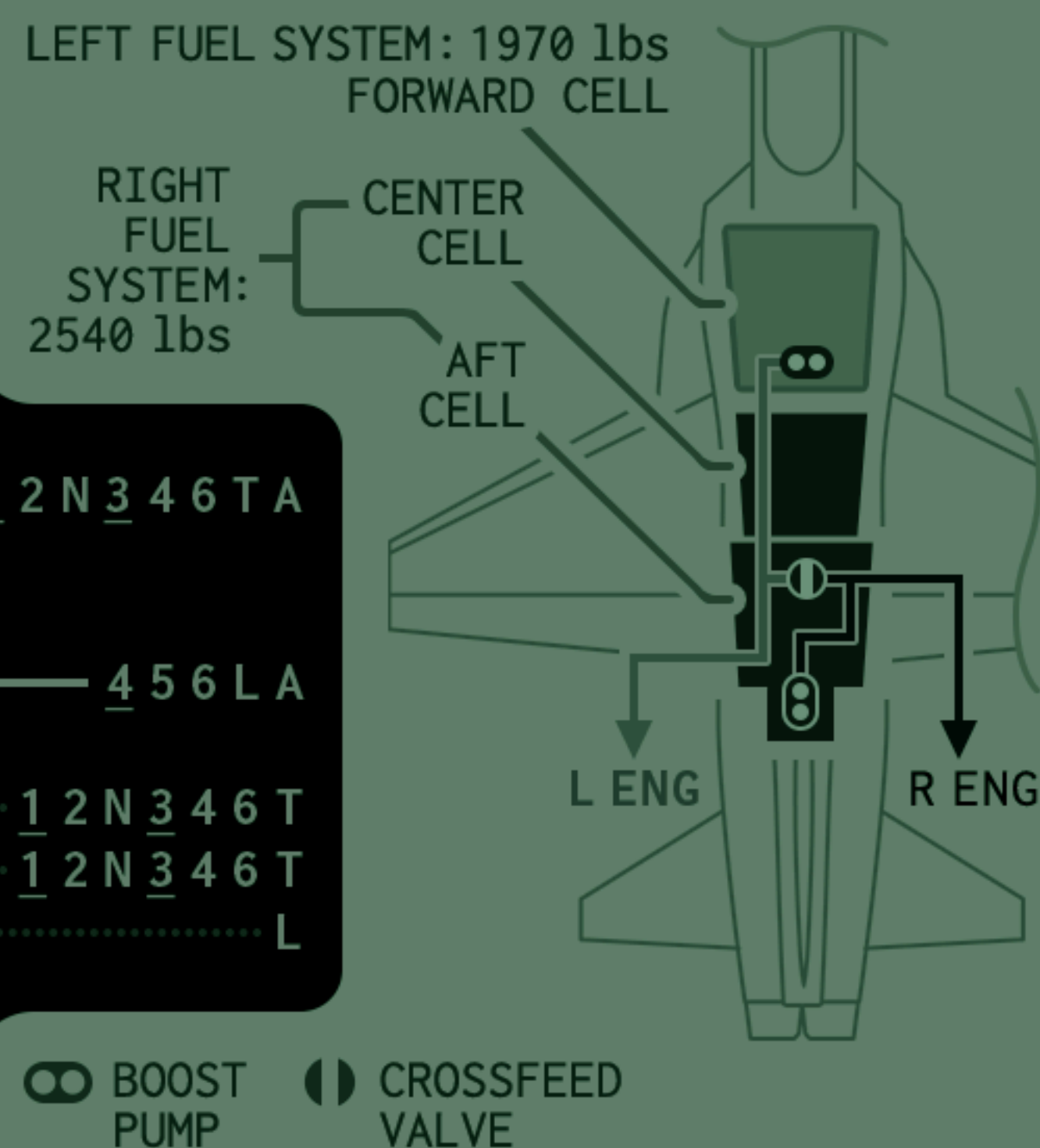
NOSE:	N	NOSE PLUGS	TAIL:	T	TAIL PLUGS	Underscored: Default
1	M904E4	mech	3	M905	mech	L: LASER CODE
2	DSU-33	prox	4	FMU-139	elec	A: COATING
			5	FMU-143	elec penetr	
			6	FMU-152	JPF	

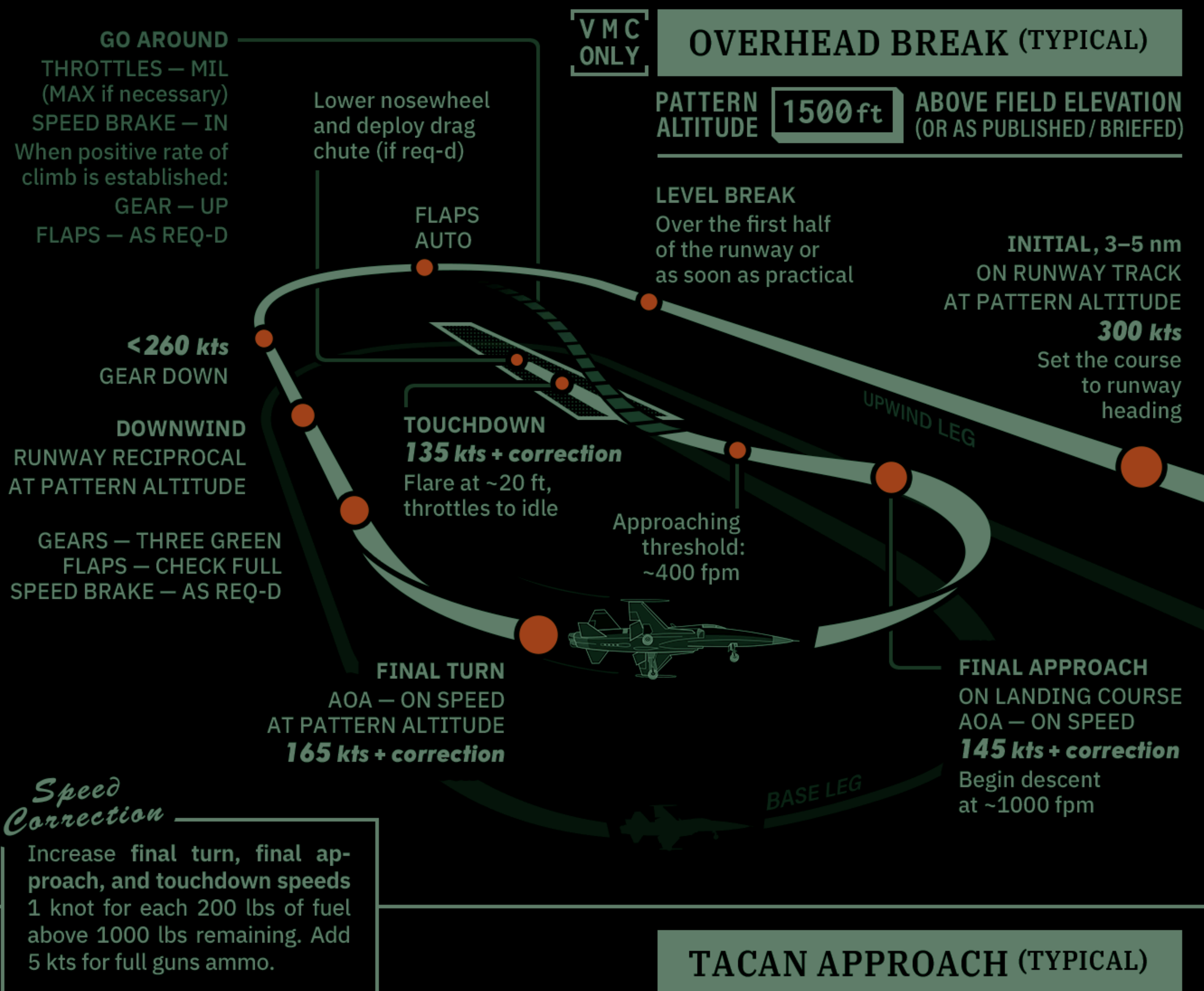
▲ FUZING OPTIONS ▼

6 5 4 3 2

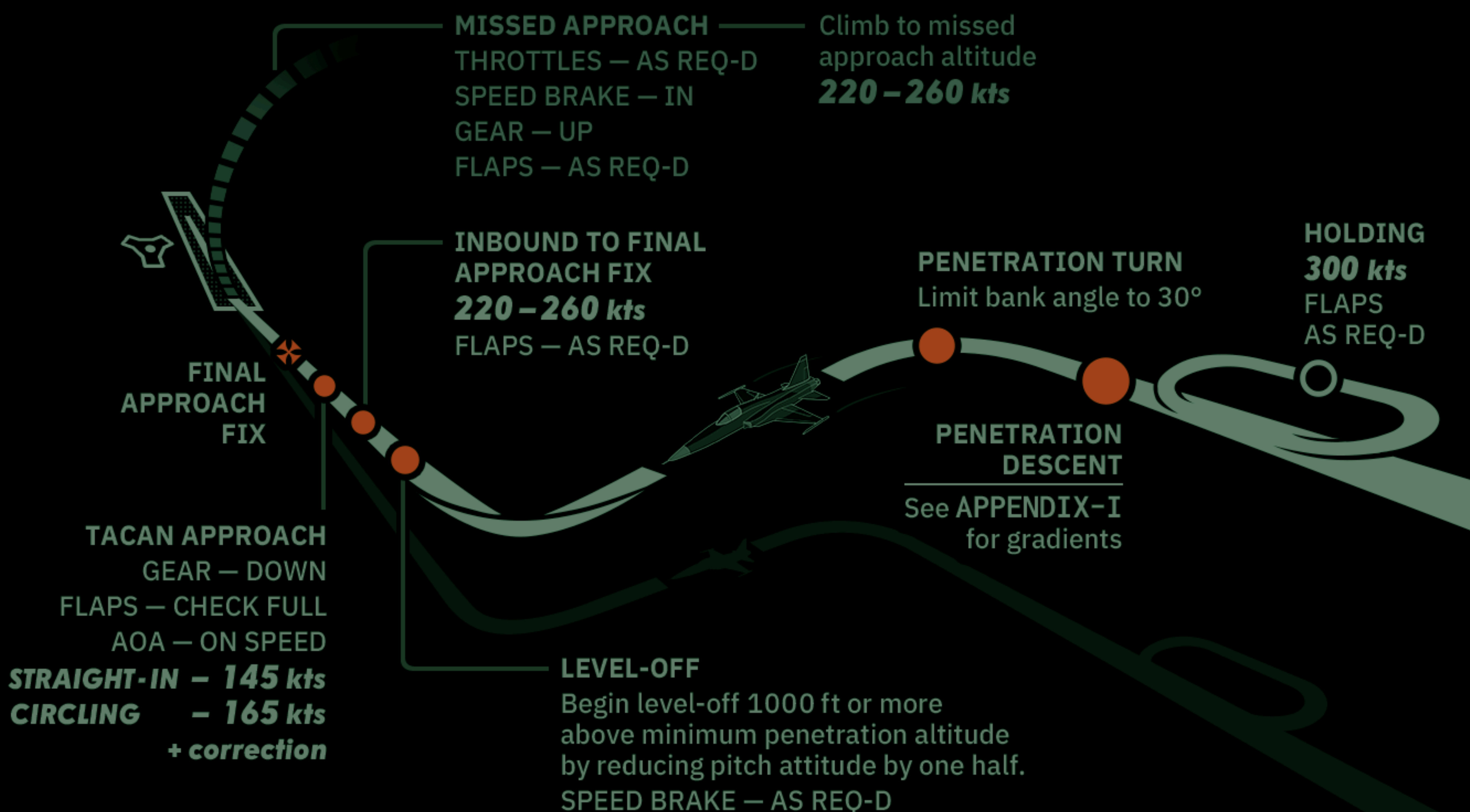
▲ WEIGHT WITHOUT PYLONS AND RACKS, LBS

1. Crossfeed Switch — CROSSFEED.
2. Boost Pump (on low fuel side) — OFF.
3. System Balanced; Disabled Boost Pump — On.
4. After at least 2 min; Crossfeed — OFF.





TACAN APPROACH (TYPICAL)



PENETRATION DESCENT GRADIENTS*

KLAS TO MAINTAIN	300	270	300	270	250	250
RPM TO MAINTAIN	80%	IDLE	80%	IDLE	IDLE	80%
FLAPS	FXD	UP	FXD	FXD	FXD	FXD
SPEED BRAKE	IN	IN	OUT	OUT	IN	OUT
REFERENCE VS, fpm	2500	5000	4500	6000+	3500	3000
DISTANCE FOR EACH 1000 feet	2-3 nm	1.2 nm	1.3 nm	0.9 nm	1.3 nm	2.0 nm

RECOMMENDED DESCENT RATES ON THE GLIDE PATH				
FUEL, lbs	< 3700		> 3700	
X-WIND	No	Yes	No	Yes
VS, fpm	600	400	400	300

MAXIMUM CROSSWIND COMPONENT DURING LANDING		
GROSS WEIGHT, lbs	W/DRAG CHUTE	W/O DRAG CHUTE
BELOW 15,500	20 kts	35 kts
ABOVE 15,500	25 kts	35 kts

* Conditions:
GW 14,000 lbs,
2000 lbs of fuel,
no stores,
no wind.

Distances vary
with drag index,
ground speed,
and wind.

HEADWIND COMPONENT



CROSSWIND COMPONENT

WIND COMPONENTS

Angle between wind direction and runway heading											
	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	
5	5 0	5 1	5 2	5 3	4 4	3 4	3 5	2 5	1 5	0 5	
10	10 0	10 2	9 3	9 5	8 6	6 8	5 9	3 9	2 10	0 10	
15	15 0	15 3	14 5	12 7	11 10	9 12	7 13	5 14	3 15	0 15	
20	20 0	20 3	19 7	17 10	15 13	13 15	10 17	7 19	3 20	0 20	
25	25 0	25 4	24 9	21 13	19 16	16 19	13 22	9 24	4 25	0 25	
30	30 0	30 5	28 10	26 15	23 19	19 23	15 26	10 28	5 30	0 30	
35	35 0	35 6	33 12	30 18	26 23	23 27	18 30	12 33	6 35	0 35	
40	40 0	39 7	38 14	35 20	31 26	26 31	20 35	14 38	7 39	0 40	
45	45 0	44 8	42 15	39 23	35 29	28 35	23 39	15 43	8 45	0 45	
50	50 0	49 9	47 17	43 25	38 32	32 38	25 43	17 47	9 49	0 50	
55	55 0	54 9	52 19	47 28	42 35	35 43	28 48	19 52	9 54	0 55	
60	60 0	59 10	56 21	52 30	46 39	39 46	30 52	21 56	10 59	0 60	
65	65 0	64 11	61 22	56 33	50 42	42 50	33 57	23 62	11 64	0 65	
70	70 0	69 12	66 24	61 35	54 45	45 54	35 61	24 66	12 69	0 70	

WIND SPEED, kts

IMPROMPTU VOR NAVIGATION

TACAN DME/VOR PAIRING TABLE		109.30	30X	110.75	44Y	112.20	59X	113.65	83Y	115.10	98X	116.55	112Y
		109.35	30Y	110.80	45X	112.25	59Y	113.70	84X	115.15	98Y	116.60	113X
		109.40	31X	110.85	45Y	112.30	70X	113.75	84Y	115.20	99X	116.65	113Y
108.00	17X	109.45	31Y	110.90	46X	112.35	70Y	113.80	85X	115.25	99Y	116.70	114X
108.05	17Y	109.50	32X	110.95	46Y	112.40	71X	113.85	85Y	115.30	100X	116.75	114Y
108.10	18X	109.55	32Y	111.00	47X	112.45	71Y	113.90	86X	115.35	100Y	116.80	115X
108.15	18Y	109.60	33X	111.05	47Y	112.50	72X	113.95	86Y	115.40	101X	116.85	115Y
108.20	19X	109.65	33Y	111.10	48X	112.55	72Y	114.00	87X	115.45	101Y	116.90	116X
108.25	19Y	109.70	34X	111.15	48Y	112.60	73X	114.05	87Y	115.50	102X	116.95	116Y
108.30	20X	109.75	34Y	111.20	49X	112.65	73Y	114.10	88X	115.55	102Y	117.00	117X
108.35	20Y	109.80	35X	111.25	49Y	112.70	74X	114.15	88Y	115.60	103X	117.05	117Y
108.40	21X	109.85	35Y	111.30	50X	112.75	74Y	114.20	89X	115.65	103Y	117.10	118X
108.45	21Y	109.90	36X	111.35	50Y	112.80	75X	114.25	89Y	115.70	104X	117.15	118Y
108.50	22X	109.95	36Y	111.40	51X	112.85	75Y	114.30	90X	115.75	104Y	117.20	119X
108.55	22Y	110.00	37X	111.45	51Y	112.90	76X	114.35	90Y	115.80	105X	117.25	119Y
108.60	23X	110.05	37Y	111.50	52X	112.95	76Y	114.40	91X	115.85	105Y	117.30	120X
108.65	23Y	110.10	38X	111.55	52Y	113.00	77X	114.45	91Y	115.90	106X	117.35	120Y
108.70	24X	110.15	38Y	111.60	53X	113.05	77Y	114.50	92X	115.95	106Y	117.40	121X
108.75	24Y	110.20	39X	111.65	53Y	113.10	78X	114.55	92Y	116.00	107X	117.45	121Y
108.80	25X	110.25	39Y	111.70	54X	113.15	78Y	114.60	93X	116.05	107Y	117.50	122X
108.85	25Y	110.30	40X	111.75	54Y	113.20	79X	114.65	93Y	116.10	108X	117.55	122Y
108.90	26X	110.35	40Y	111.80	55X	113.25	79Y	114.70	94X	116.15	108Y	117.60	123X
108.95	26Y	110.40	41X	111.85	55Y	113.30	80X	114.75	94Y	116.20	109X	117.65	123Y
109.00	27X	110.45	41Y	111.90	56X	113.35	80Y	114.80	95X	116.25	109Y	117.70	124X
109.05	27Y	110.50	42X	111.95	56Y	113.40	81X	114.85	95Y	116.30	110X	117.75	124Y
109.10	28X	110.55	42Y	112.00	57X	113.45	81Y	114.90	96X	116.35	110Y	117.80	125X
109.15	28Y	110.60	43X	112.05	57Y	113.50	82X	114.95	96Y	116.40	111X	117.85	125Y
109.20	29X	110.65	43Y	112.10	58X	113.55	82Y	115.00	97X	116.45	111Y	117.90	126X
109.25	29Y	110.70	44X	112.15	58Y	113.60	83X	115.05	97Y	116.50	112X	117.95	126Y

Set the UHF radio to ADF mode and tune it to a VOR frequency by using T instead of the leading 1. Set the TACAN to T/R mode and tune it to a VORDME channel equivalent from the table above.

Switch between the DF and TACAN NAV MODES as desired to display bearing or distance to the VORDME beacon on the HSI.

Switch the UHF back to BOTH whenever needed to talk to ATC.

CIRCUIT BREAKER PANELS

ARMAMENT		115/200 VOLT AC				28 VOLT DC	
LEFT AIM-9 POWER (5)	LEFT GUN FIRING (5)	28 VOLT AC POWER (1½)	ATTD & HDG REF SYS (5)	CENTRAL AIR DATA COMPUTER (5)	ENG IGN L ENG INST & HYD IND (5)	PYLON TANK FUEL CONT (5)	UHF CMND RADIO (5)
						L BOOST CL & TIP TANK FUEL CONT (5)	LEFT LE FLAP CONT (5)
RIGHT AIM-9 POWER (5)	RIGHT GUN FIRING (5)	TRIM CONTROL (5)	ATTD & HDG REF SYS (5)	TRIM CONTROL (5)	L ENG AUX DOOR (5)	IGNITION INVERTER POWER (10)	RIGHT LE FLAP CONT (5)
						L ENG START & AB CONT (5)	LEFT TE FLAP CONT (5)
CABIN COND (5)	FLT DIR COMPUTER (5)	FUEL QTY - PRIMARY (5)	ATTD & HDG REF SYS (5)	TACAN (5)	DIRECTIONAL GYRO (5)	R ENG START & AB CONT (5)	RIGHT TE FLAP CONT & IND (5)
LEFT		28 VOLT DC		PEDESTAL		115/200 VOLT AC	
RIGHT							
WPN PWR LEFT INBD (7½)	WPN PWR CENTER LINE (7½)	WPN PWR RIGHT INBD (7½)	PITOT HEATER (7½)		R OIL & HYD IND FUEL QTY SEL (1)	CABIN AIR VALVES (5)	
WPN PWR LEFT OUTBD (7½)	WPN ARMING (7½)	WPN PWR RIGHT OUTBD (7½)		INST LIGHTS (5)	R ENG AUX DOOR (5)	BLANKING ELECTRONIC UNIT (5)	
JETTISON CONTROL (5)	WPN RELEASE (7½)	WPN MODE SEL & AIM-9-INTLK (7½)					
EMERGENCY ALL JETTISON (15)	LEFT AIM-9 CONT (7½)	RIGHT AIM-9 CONT (7½)		CAUTION & WARN LIGHTS - DIM (5)	QXY QTY & CANOPY SEAL (5)	LDG - TAXI LAMP PWR (5)	



NOTE ON RADIO AND ADF

While nominally limited to 225–399 MHz, the ‘ARC–164’ radio in DCS de-facto supports 100.000–399.975 MHz range.

Consequently, the ARA–50 in ADF + DF mode can display bearing to any signal in that range (voice, VOR, ILS, etc).

To tune to 1XX.XXX A frequencies, rotate the first radio dial knob to the T position.

LANDING LIGHTS OPERATION

If the position lights (the NAV knob) are on, the landing lights automatically extend and retract together with the landing gear.

The LDG & TAXI switch is used to turn the extended landing lights on and off.

Lights automatically switch to low intensity while on the ground.

Landing lights can stuck in the extended position if gear is retracted with position lights turned off. Cycle the NAV knob to resolve.

THE MODULE'S MISSING FEATURES

Consider the following when planning missions:

- Fire warnings, nozzle damage, and canopy cracks are purely cosmetic and have no impact on flying.
- Engine mechanical failures are not implemented.
- Battery cannot be discharged or overloaded.
- Icing is not simulated.
- SAI can operate on battery power indefinitely.
- ‘DC OVERLOAD’, ‘DIR GYRO’, ‘IFF’, and ‘INS’ cautions are not simulated.

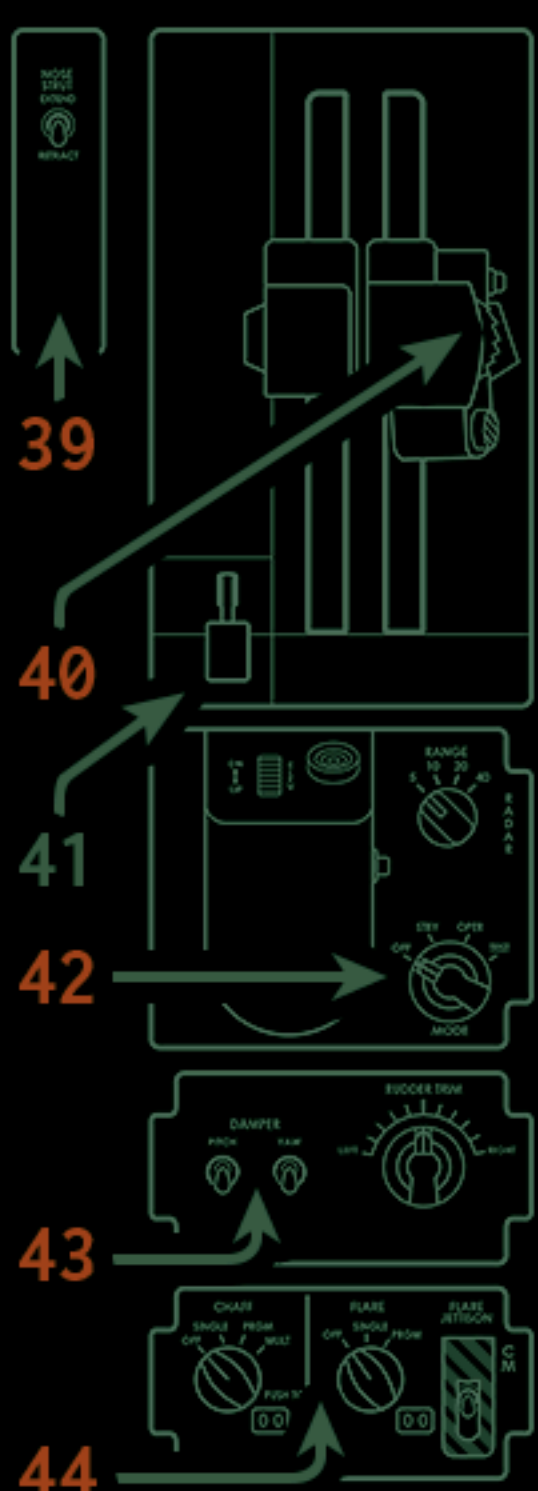
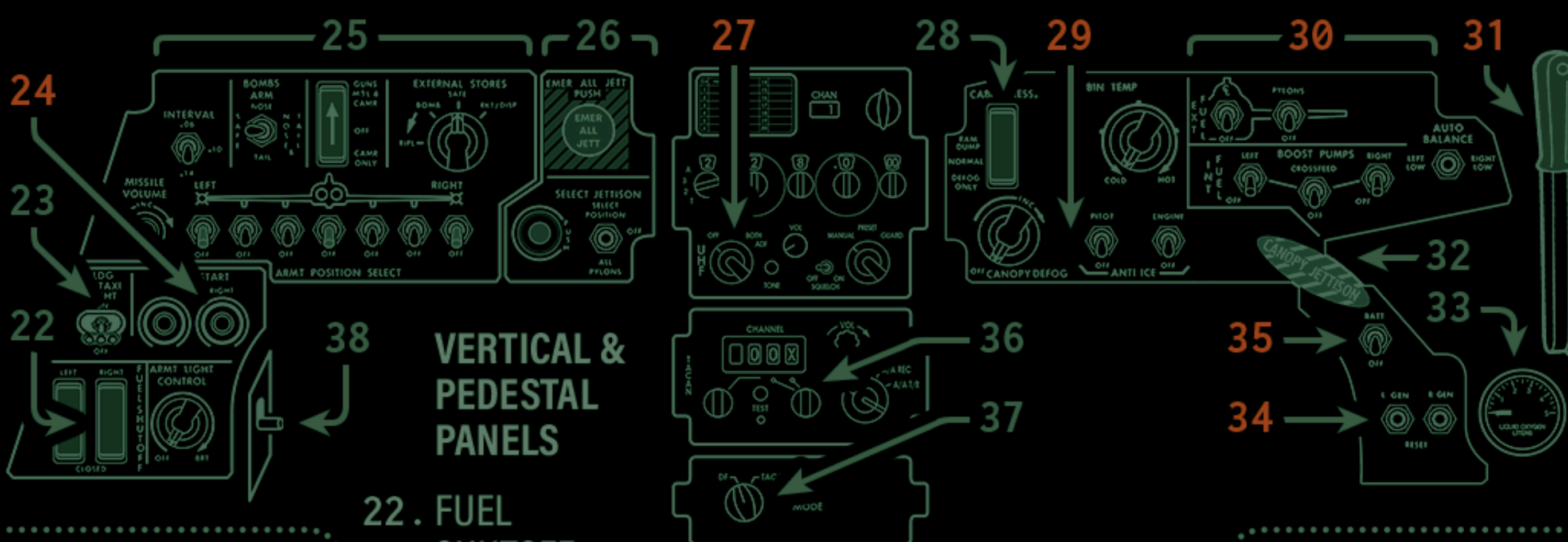
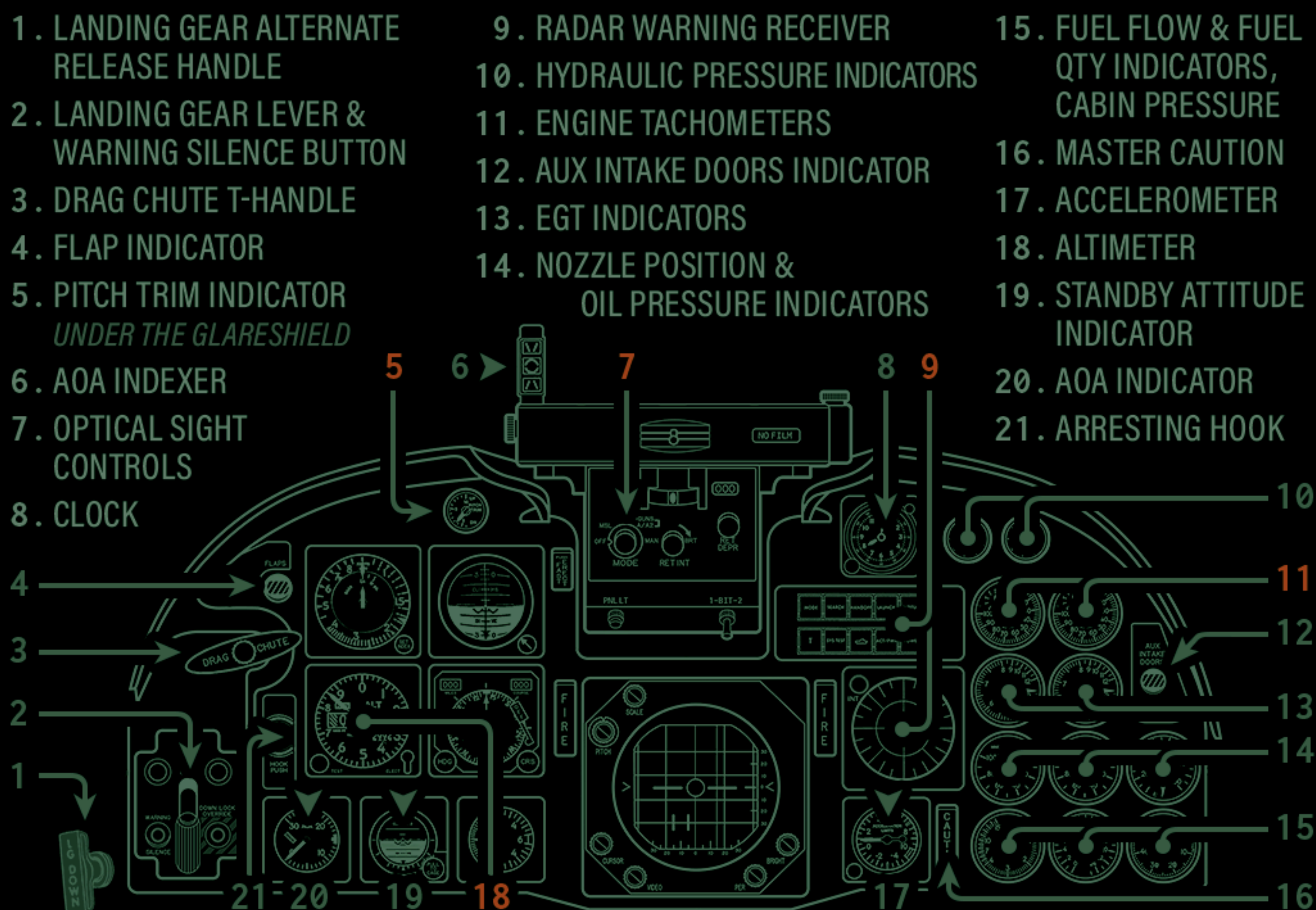
The following **emergency procedures** are non-applicable and were not included in these kneeboards: ‘Nosewheel Shimmy’, ‘Electrical Fire’, ‘Smoke in Cockpit’, ‘Engine Malfunctions’, ‘Airframe Gearbox Failure’, ‘Ditching’.

The following **failure triggers** have no effect: ‘Fire Engine’, ‘Gearbox’, ‘Hydraulic System’, ‘Low Pressure Leakage’, ‘Rectifier’, ‘Nozzle Control System’, ‘Oil System’, ‘Oxygen System Total Failure’.

PROHIBITED MANEUVERS

- **Sustained 0-G or negative-G flight** at high engine power settings. May lead to engine fuel starvation and flameout within 10-120 seconds, depending on altitude and how much fuel flow exceeds 650 lb/h.
- **Flying in icing conditions.** When icing conditions are anticipated, turn on the pitot heat, engine anti-ice, and canopy defog. Maintain at least 80% RPM to ensure effective engine anti-icing. Leave the area of icing conditions as soon as possible.
- **Flying through thunderstorms or extreme turbulence.** If cannot be avoided, the best penetration speed is 300 kts. Trim for level flight. Do not change thrust and trim. Maintain attitude. Do not chase altitude and vertical velocity indicators.
- **Exceeding aileron spring stop** except for spin recovery and emergencies. (*Structural damage is not simulated.*)

INSTRUMENT PANEL



LEFT & RIGHT CONSOLES

QUICKSTART

- 27 UHF BOTH (rmb)
- 30 BOOST PUMPS LEFT & RIGHT
- 35 BATT
- 34 L GEN & R GEN (rmb)

Ground Air Supply:
Connect & Apply

- 11 At >10% Left RPM:
- 24 ENGINE START LEFT

Left Throttle: IDLE
(RAIt + Home)

- 11 Wait for Left RPM to stabilize at ~50% (in about 30 sec)

Ground Air Supply:
Apply

- 11 At >10% Right RPM:
- 24 ENGINE START RIGHT

Right Throttle: IDLE
(RShift + Home)

- 11 Wait for Right RPM to stabilize at ~50%

Ground Air Supply:
Disconnect

- 31 Close (LCtrl+C)

- 40 Speed Brake IN (fwd) (rmb)

- 40 Flap Thumb Switch AUTO (aft)

- 43 PITCH & YAW

- 18 Set QNH

- 5 Set Pitch Trim based on loadout and note rotation speed for takeoff:

CLEAN: 7 / 150 kts
MEDIUM: 8 / 165 kts
HEAVY: 9 / 180 kts

BEFORE TAKEOFF:

- 39 Extend Strut (fwd)

- 7 Sight as desired

- 9 RWR POWER

- 42 Radar OPER

- 44 CMs as desired

- 29 PITOT

GENERAL		INFLIGHT (continued)	
AOA/FLAPS FAILURE	1	ELECTRICAL SYSTEM FAILURE	15
CADC/PITOT-STATIC MALFUNCTION	2	ENGINE FAILURE IN FLIGHT /	
STATIC INVERTER FAILURE	3	ENGINE FAILURE AT LOW ALTITUDE	16
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CAUTION LIGHT	CONDITION	CORRECTIVE ACTION	
FIRE WARNING	Engine fire or overheat.	See FIRE WARNING procedures.	D T
AIR DATA COMPUTER	CADC outputs unreliable.	See CADC/PITOT-STATIC MALFUNCTION /2.	D T
AOA / FLAPS	AOA switching unit failure.	See AOA/FLAPS FAILURE /1.	D T
CANOPY	A) Canopy unlocked. B) Canopy lost or jettisoned.	A) Lock canopy. B) See LOSS OF CANOPY /21.	D
ENGINE ANTI-ICE ON	Anti-ice system operating.	Advisory.	
EXT TANKS EMPTY	External tanks empty.	Advisory.	
FLIGHT HYD UTILITY HYD	Hydraulic pressure in indicated system 1500 psi or less.	See HYDRAULIC SYSTEMS FAILURE /20.	D
L FUEL LOW R FUEL LOW	Usable fuel remaining in indicated system 400 lbs or less.	Check fuel balance.	
L FUEL PRESS R FUEL PRESS	Fuel pressure in indicated system 6.5 psi or less.	Check boost pumps ON, reduce RPM, descend below 25,000 feet. Monitor fuel gages. See FUEL LEAK /19 if a fuel leak is suspected.	D T
L GENERATOR R GENERATOR	Indicated generator not operating.	Turn ON or RESET generator. See ELECTRICAL SYSTEM FAILURE /15.	D T
OXYGEN	Oxygen remaining 0,5 liter or less, or pressure 40 psi or less.	Descend to a safe altitude, monitor supply pressure.	

BOLDFACE TEXT: critical steps performed immediately from memory.
D)amage, T)rigger: possible failure causes.



GENERAL

AOA / FLAPS FAILURE ^{DT}

1

Symptom: AOA/FLAPS caution.

Effect: Loss of AUTO flaps operation. Flaps remain in the position attained at the time of failure until another setting is selected.

1. Flap Thumb Switch — FXD or UP.
2. (Approach and landing) Flap Lever — FULL.

CADC / PITOT STATIC ^{DT}
MALFUNCTION

2

Symptoms:

AIR DATA COMPUTER caution — CADC failure.
Loss of airspeed gauge — pitot-static leakage.

Effects of CADC failure:

Loss of altimeter primary ELECT mode.
Loss of AUTO and FXD flap modes.
Loss of landing gear warning.

Effects of pitot-static malfunction:

Loss of airspeed gauge.
Loss of aux intake doors control.
Inadequate landing gear warnings.

1. Altimeter — Standby PNEU Mode.
2. Flap Thumb Switch — UP.
3. (Approach and landing) Flap Lever — FULL.
4. Aux Door circuit breakers — As Desired.

Failed pitot-static causes aux doors to open.
Pulling L ENG AUX DOOR and R ENG AUX DOOR circuit breakers closes doors and prevents their cycling that may lead to unexpected loss of thrust.

STATIC INVERTER FAILURE ^{DT}

3

Symptom: Failure of fuel and oxygen indicators to respond during GAGE TEST & QTY CHECK.

Effect: Engine ignition not available when aircraft is on battery power.

Starting an engine with failed static inverter requires an operating generator or connection of Ground Electric Power.

Avoid maneuvers that may lead to dual engine flameout, as airstart will not be possible.

GROUND OPERATIONS

EMERGENCY EXIT ON GROUND

4

1. Canopy — Open or jettison.
2. Throttles — OFF.

ENGINE FIRE DURING START ^{DT}

5

1. Throttles — OFF.
If engines fail to shut down:
2. Fuel Shutoff Switches — CLOSED.

TAKEOFF

ABORT TAKEOFF

6

1. **THROTTLES — IDLE.**
2. **CHUTE — DEPLOY.**
3. **WHEEL BRAKES — APPLY MAXIMUM.**
If arrestment need is evident:
4. **HOOK — DOWN.**

EMERGENCY JETTISON

7

1. EMER ALL JETT Button — Push.
If stores fail to jettison:
2. SELECT JETTISON Switch — ALL PYLONS.
3. SELECT JETTISON Button — Push.

ENGINE FAILURE/FIRE
WARNING DURING TAKEOFF ^{DT}

8

If takeoff is refused:

1. Abort **/6**.
If engine fire is suspected, place the throttle of the affected engine to OFF.
If the fire is confirmed, perform Emergency Exit On Ground **/4** procedure after stopping.
- If takeoff is continued:
1. **THROTTLES — MAX.**
2. **STORES — JETTISON (IF NECESSARY) /7.**
3. At safe ejection altitude (2000 ft AGL) — Perform ENGINE FAILURE IN FLIGHT **/16** and/or **FIRE WARNING** procedures.

LANDING GEAR ^D
RETRACTION FAILURE

9

Retraction of the main gear requires at least 1500 psi in the utility hydraulic system.

If the landing gear warning light remains on after lever has been moved to LG UP:

- 1. **Airspeed — Maintain below 260 kts.**
- 2. **Gear Alternate Release Reset Control — Cycle.**
- 3. **Nose Strut — Retracted (switch in aft pos).**
- 4. **Gear Lever — LG DOWN, Then LG UP.**
- 5. **Throttles — cycle to MIL.**
- 6. **If light remains on — Lower gear.**

If light goes out, proceed with flight.

Note that a damaged gear may fail to retract, but its indicator light may still go out.

SINGLE ENGINE TAKEOFF

10

If an engine fails and takeoff is continued:

- Use MAX thrust until reaching the safe ejection altitude (2000 ft AGL).
- Jettison stores if acceleration is inadequate.
- Use remaining runway to increase airspeed as much as possible.
- Immediately after liftoff, focus on accelerating, not on gaining altitude, until confident that the aircraft can maintain a minimum climb rate of 300 fpm.
- Raise landing gear after a steady rate of climb is attained, but no later than 210 kts.
- Recommended single-engine climb speeds:

GEAR DOWN, FLAPS AUTO - 210 kts
GEAR UP, FLAPS AUTO - 230 kts
GEAR UP, FLAPS UP - 290 kts

TIRE FAILURE ON TAKEOFF ^D

11

If takeoff is refused:

- 1. **Abort /6.**

If takeoff is continued:

- 1. **GEAR — DO NOT RETRACT.**
- 2. See **LANDING WITH TIRE FAILURE /30.**

AIRSTART / MAXIMUM GLIDE

12

- | If both engines flame out with inoperative static inverter /3, restart is not possible.
- | **Frozen (0% RPM) engine cannot be restarted.**

Procedure applies to affected engine. If both engines flame out, left engine start should be attempted first.

- 1. **Throttle — OFF.**
- 2. **Altitude — Below 30,000 feet.**
- 3. **Airspeed — 250 kts (optimal).**
- 4. **Boost Pump Switches — LEFT and RIGHT.**
- 5. **Battery Switch — Check BATT.**
- 6. **Start Button — PUSH.**
- 7. **Throttle — IDLE.**

If restart does not occur within 50 seconds, increase engine windmill speed by increasing airspeed and/or decreasing altitude before attempting another start.

■ ALTERNATE AIRSTART

- 1. **Altitude — Below 30,000 feet.**
- 2. **Airspeed — 250 kts (optimal).**
- 3. **Battery Switch — Check BATT.**
- 4. **Throttle — Advance to MAX.**
This activates ignition. If the throttle is already at MAX, cycle it to MIL and return to MAX.

■ MAXIMUM GLIDE (BOTH ENGINES WINDMILLING)

- 1. **Flaps — Up.**
- 2. **Stores — Jettisoned.**
- 3. **Airspeed — 250 kts (optimal for airstart).**

Approximate glide distance for each 1000 ft AGL, with or without pylons: 1 nm.



CONTROLLABILITY CHECK13

If the aircraft's normal performance has been degraded by damage, failures, or other factors:

1. **Altitude** — 15,000 feet AGL (if practical).
2. **Landing Configuration** — Establish.
Minimize flap movement if flap damage is known or suspected.
3. **Airspeed** — Reduce to determine the acceptable approach and landing characteristics.

■ APPROACH AND LANDING

1. **Gear and Flaps** — As determined.
Do not change landing configuration.
2. **Landing Approach** — Straight In.
Plan to fly a power-on, straight in approach requiring minimum flare.
3. **Landing Speed** — No slower than determined airspeed.
4. **Touchdown** — At or above determined airspeed.

EJECTION14

- Ejection is preferable to landing on an unprepared surface. Do not attempt landing with both engines flamed-out.
- Eject before the start of any sink rate.
- If possible, eject above 2000 ft AGL flying straight and level at low airspeed.
- When below 2000 ft AGL, zoom to trade speed for altitude, and eject before climb rate reaches zero.
- Under uncontrolled conditions (spins, dives) eject at least 10,000 ft AGL whenever possible.

■ BEFORE EJECTION

1. IFF to EMER. UHF to GUARD. Transmit mayday, giving position and intentions.
2. Turn aircraft toward uninhabited area.
3. Wings level. Slow down as much as practical. Attempt a positive rate-of-climb.

ELECTRICAL SYSTEM FAILURE DT15

■ ALTERNATE AIRSTART

Symptom: L or R GENERATOR caution.

Effect: Loss of control over the corresponding aux intake door (door remains closed).

The remaining generator picks up the load.

Note | If one engine is frozen (0% RPM), the remaining engine requires RPM above idle (approx. 68%) to keep generator on line.

■ AC / COMPLETE ELECTRICAL FAILURE

Symptom: L and R GENERATOR cautions.

Effect: The following are inoperative with complete electrical failure:

- FLIGHT AND ENGINE INSTRUMENTS

EXCEPTION:

- TACHOMETERS
- AIRSPEED INDICATOR
- ALTIMETER IN PNEU MODE
- STANDBY ATTITUDE INDICATOR
- VERTICAL VELOCITY INDICATOR
- ACCELEROMETER
- ARRESTING HOOK EXTENSION
- AUX INTAKE DOORS
- FLAPS
- FUEL BOOST PUMPS
- OPTICAL SIGHT
- PITCH AND AILERON TRIM
- RADAR
- RADAR WARNING RECEIVER
- STABILITY AUGMENTER SYSTEM
- TACAN, ADF

1. **Throttle(s)** — Retard (if above 25,000 ft).
2. **Altitude** — 25,000 ft or below.
3. **Battery Switch** — Check BATT.
4. **Generator Switch(es)** — Reset, then L GEN/ R GEN.
5. **Crossfeed** — As Req.
6. **Altimeter** — Standby PNEU Mode.
7. **Land as soon as possible.**

Operate only systems necessary for flight and landing.

INFLIGHT (continued)

ENGINE FAILURE IN FLIGHT /
ENGINE FAILURE AT LOW ALTITUDE

16

Effects of RIGHT engine failure: Loss of right generator and flight control hydraulic pump.

Left generator picks up the load. Utility system provides hydraulic power for the flight controls.

Effects of LEFT engine failure: Loss of left generator and utility hydraulic pump. Any flight control movement rapidly depletes hydraulic pressure in the utility system.

Right generator picks up the load. Some hydraulic pressure is regained from windmilling RPM, but at a rate insufficient for reliable operation of the following:

- LANDING GEAR NORMAL EXTENSION
- NOSEWHEEL STEERING
- SPEED BRAKE
- STABILITY AUGMENTER SYSTEM
- WHEEL BRAKES

ENGINE FAILURE IN FLIGHT

1. **Throttle (good engine) — As Req.**
2. **Stores — Jettison (if necessary).**
3. **Gear / Speed Brake — Up / In.**
4. **Throttle (failed engine) — OFF.**
5. **Fuel Balancing — As Req.**
Autobalance should be used, if available.
With fuel less than 400 lbs in each system:
6. **Fuel Boost Pumps — LEFT and RIGHT.**
7. **Crossfeed Switch — CROSSFEED.**

ENGINE FAILURE AT LOW ALTITUDE

1. **THROTTLES — MAX.**
2. **Engine Instruments — Monitor.**
If both engines fail at low altitude and with sufficient airspeed:
 - Zoom to trade speed for altitude.
 - Try to airstart immediately /12.
 - Ejection should be done while in a positive climb.
 - If continued airstarts are to be attempted, lower the nose before airspeed drops below 250 kts.

FIRE WARNING IN FLIGHT ^{DT}
(AFFECTED ENGINE)

17

1. **THROTTLE — IDLE.**
2. **THROTTLE — OFF IF FIRE WARNING LIGHT REMAINS ON.**
Close fuel shutoff switch if engine fails to shut down or if fire warning light remains on.
3. **IF FIRE IS CONFIRMED — EJECT /14.**

FUEL AUTOBALANCE
SYSTEM MALFUNCTION ^{DT}

18

AUTO BALANCE SWITCH STUCK
IN CENTER POSITION

Autobalance system is considered inoperative.

1. **Use manual balancing (as required).**

AUTO BALANCE SWITCH STUCK
IN LEFT LOW OR RIGHT LOW POSITION

With this failure, the autobalance system continuously draws fuel from the selected fuel system until it is empty. After that, the fuel from the opposite system is drawn by gravity flow.

This procedure puts both fuel systems in gravity feed mode after their fuel quantity has been evened by autobalance.

1. **Fuel Quantity — Check often.**
Once both systems are balanced:
2. **Altitude — 25,000 ft or below.**
3. **Crossfeed Switch — CROSSFEED.**
4. **Fuel Boost Pump on the side opposite to where the Auto Balance switch is stuck — OFF.**

L FUEL PRESS and R FUEL PRESS caution lights indicate gravity feed from both fuel systems.

5. **Land as soon as practical.**
Fly with reduced power at lowest practical altitude to assure continued stable engine operation with boost pumps inoperative.
If required, perform external fuel transfer.

FUEL LEAK ^D

19

Damage to the internal fuel tanks can result in a rapid and evident decrease in fuel quantity in one or both fuel systems.

Once occurred, fuel leaks cannot be stopped or reduced. Depending on the severity of the damage, the system may lose the remaining fuel and cause engine flameout in less than one minute.

Because fuel leaks don't have dedicated warnings, it is important to check fuel quantity each time:

- Note
- After egress/fencing out;
 - When the aircraft sustains damage;
 - When the FUEL PRESS caution illuminates.

■ IMMEDIATELY UPON DETECTION OF RAPID FUEL LOSS

1. **THROTTLES — MAX.**
2. **CLIMB ABOVE SAFE EJECTION ALTITUDE.**
3. **DETERMINE THE AFFECTED FUEL SYSTEM.**

■ BOTH SYSTEMS AFFECTED

1. Trade remaining fuel for altitude and time.
2. Prepare for controlled ejection after both engines flame out.

If speed and altitude permit, an airstart using fuel from the external tanks may be attempted as a last resort to save the aircraft.

■ ONE SYSTEM AFFECTED

1. Boost Pump on the affected side — OFF.
2. Crossfeed Switch — CROSSFEED.
This might prevent engine flameout when the affected system runs out of fuel.
3. Throttles — As required to maintain at least 250 kts.
4. Climb to a safe altitude, but below 25,000 ft.
5. Land as soon as practical.
If needed, an airstart and/or fuel transfer from the external tanks may be attempted.

HYDRAULIC SYSTEMS FAILURE ^D

20

Hydraulic system failure is indicated by a total and irreversible loss of hydraulic pressure in the affected system.

Note

It should not be confused with pressure fluctuations caused by the loss of hydraulic pump following engine failure.

■ DUAL SYSTEM FAILURE

Symptoms: FLIGHT HYD and UTILITY HYD cautions. Loss of pressure in both systems.

1. If flight control becomes impossible — Eject.

■ FLIGHT CONTROL SYSTEM FAILURE

Symptoms: FLIGHT HYD caution. Loss of pressure in the flight control system.

The utility system has enough capacity to provide hydraulic power for the flight controls.

1. Land as soon as practical.

■ UTILITY SYSTEM FAILURE

Symptoms: UTILITY HYD caution. Loss of pressure in the utility system.

Effect: The following are inoperative with utility system failure:

- GUN GAS DEFLECTOR AND BAY PURGE DOORS
- LANDING GEAR NORMAL EXTENSION
- NOSEWHEEL STEERING
- SPEED BRAKE
- STABILITY AUGMENTER SYSTEM
- WHEEL BRAKES

1. Pitch and Yaw Damper Switches — OFF.
2. Land as soon as practical.

LOSS OF CANOPY ^D

21

If the canopy is lost or jettisoned:

1. Reduce airspeed to 300 kts or less.
2. Oxygen Supply Lever — ON.

If the oxygen supply is not available, descend to altitude not requiring oxygen.

INFLIGHT (continued)

PITCH DAMPER FAILURE ^T

22

Symptoms: Applying aft or forward stick results in excessive oscillations in pitch that take longer than usual to settle.

1. Pitch Damper Switch — Verify in PITCH.
2. (When carrying external tanks) Airspeed — Reduce below 0,75 M.

PITCH TRIM FAILURE ^T

23

1. Airspeed — Adjust (as necessary to minimize stick forces).
2. Controllability check — Perform /13. Approach and landing — perform as instructed.

SPIN & POST-STALL GYRATION RECOVERY

24

■ ERECT PSG RECOVERY

When uncontrolled aircraft oscillations at angles of attack above stall are encountered:

1. Relax aft stick pressure to reduce AOA.
If this does not immediately resolve PSG:
2. **STICK — FORWARD AS REQUIRED.**
Delayed application may result in spin entry.
3. Ailerons and Rudder — Neutral.

Recovery is indicated when the aircraft responds to forward stick and airspeed exceeds 130 kts.

4. Relax forward stick after recovery. Smoothly apply aft stick to regain positive G flight.
Continued application of forward stick may result in inverted PSG or inverted spin entry. Failure to apply aft stick may result in the inverted pitch hangup (IPH).

PSG recovery may require 4000 feet of altitude, and approx. 6000 feet for dive pullout.

■ ERECT SPIN RECOVERY

Immediately upon recognition of spin direction:

1. **STICK — FULL FORWARD.**
2. **AILERON — FULL IN DIRECTION OF SPIN (bypass the Aileron Limiter).**
3. **RUDDER — FULL OPPOSITE.**

4. Flaps — AUTO.
Spin recovery begins with gradual increase of nose-low attitude, and ends in a strong pitchover when airspeed increases through 130 kts.
5. Neutralize controls after recovery. Smoothly apply aft stick to regain positive G flight.
Altitude loss per turn is approx. 1700–2500 feet. Spin recovery requires a minimum of two turns and 4000 feet of altitude. Dive pullout requires approx. 6000 feet.

■ INVERTED PSG / INVERTED SPIN RECOVERY

1. **FLAPS — UP.**
2. **STICK — AFT AS REQUIRED.**
3. **AILERONS AND RUDDER — NEUTRAL.**
Recovery is indicated by the decrease of negative G and the onset of positive G.
4. Relax aft stick after recovery.

If considerable aft stick is maintained after recovery, entry into an erect PSG or spin is possible.

Altitude loss during inverted PSG or spin may exceed 6000 feet (not including dive pullout).

Note

A windmilling engine has a very high probability of freezing (RPM drops below 10%) during stalls and spins, rendering restart impossible.

LANDING

ARRESTMENT (APPROACH END)

25

None of the airfields in the operational area are equipped with functional arresting gear. This procedure is intended for practicing.

1. Landing configuration — Establish.
2. Hook — Down.
3. AOA — On Speed.
4. Touchdown — In center of runway, minimum 500 feet before cable.
5. Chute — Deploy.
Discontinue braking before crossing cable.
6. (After engagement) Throttles — As required to control rollback.

LANDING (continued)

DRAG CHUTE FAILURE26

If the drag chute fails to deploy on landing, and the decision is made to go around:

- 1. Drag Chute — Jettison.
- 2. Go Around /See CHECKLIST–II.

SINGLE-ENGINE APPROACH & LANDING27

■ SINGLE-ENGINE APPROACH

- Delay lowering landing gear until just before glide path. MAX thrust should be used on single-engine approaches if necessary.
- Expend or jettison stores before entering the landing pattern, if necessary.
- Smoothly apply the rudder to counter yaw. Avoid abrupt control inputs, as the aircraft in this configuration is very susceptible to spin entry.

■ SINGLE-ENGINE LANDING

- 1. Flap Thumb Switch — AUTO.
- 2. Gear — Down.
With a failed left engine, windmilling RPM may be sufficient for normal gear extension that requires at least 2700 psi in the utility system. However, if gear does not extend, use the Alternate Release Handle. Wheel brakes and nosewheel steering are not available.
- 3. Airspeed — Increase 10 kts above normal until landing assured.
- 4. AOA Indicator — 14.0 units on final approach.
- 5. Drag Chute — As required.

■ SINGLE-ENGINE MISSED APPROACH

- Use MAX thrust.
- Raise landing gear after a steady rate of climb is attained.
- Keep flaps in AUTO.
- Climb at 260 kts with landing gear up or 210 kts with landing gear down.

LANDING GEAR ALTERNATE EXTENSION28

- 1. Airspeed — 260 kts or less.
- 2. Gear Lever — LG DOWN.
- 3. Gear Alternate Release Handle — Pull.
Alternate extension takes approx. 15 seconds.
- 4. Gear Indicators — Check three green.

If the main gear fails to extend fully, rocking wings or yawing the aircraft aid the extension.

If repeated extension attempts are unsuccessful, see Landing Gear Extension Failure /29.

Note Normal extension of the main gear requires at least 2700 psi in the utility system.
To resume normal operation of the Gear Lever after using Alternate Release Handle, cycle the Alternate Release Reset Control.

LANDING GEAR EXTENSION FAILURE D29

Use the following table to determine whether a landing with a particular gear condition is feasible, or ejection is the best course of action:

RECOMMENDED ACTION DEPENDING ON THE ACTUAL (NOT INDICATED) GEAR POSITION					
NOSE	UP	UP	UP	DOWN	DOWN
MAIN	BOTH DOWN	BOTH UP	ONE DOWN	BOTH UP	ONE DOWN
	LAND	EJECT	EJECT		
UNLESS LOADOUT ALLOWS BELLY LANDING					

Fly a normal landing pattern.
If possible, reduce gross weight by expending excess fuel and jettison armament.

If belly landing is to be attempted, retain empty fuel tanks on wing and/or centerline pylons to cushion shock of nose slam-down. If not possible, attain a clean no pylon configuration.

Note Gear is considered unsafe when damaged or stuck in a partially extended position. Unsafe gear may collapse at touchdown.

(Continued on the next page)

LANDING (continued)

LANDING GEAR
EXTENSION FAILURE (CONTINUED)

■ LANDING WITH NOSE GEAR UP OR UNSAFE

- 1. **Throttles** — IDLE at touchdown.
- 2. **Nose** — Gently lower to runway.
If nose gear is up, position throttles OFF when nose contacts runway.
- 3. **Drag Chute** — Deploy.
Do not use brakes if a safe stop can be made without them when the nose gear is unsafe.
- 4. **Battery Switch** — OFF.

■ BELLY LANDING

Without empty fuel tanks:

- 1. **Eject.**

With empty fuel tanks:

- 1. **Gear** — Up.
- 2. **Throttles** — OFF at touchdown.
- 3. **Drag Chute** — Deploy when on runway.
- 4. **Battery Switch** — OFF.

With no pylons installed:

- 1. **Gear** — Up.
- 2. **Speed Brake** — Open.
- 3. **Flap Thumb Switch** — AUTO.
- 4. **Airspeed** — Increase touchdown speed 10 kts.
Fly a power on approach requiring minimum flare.
- 5. **Throttles** — OFF at touchdown.
- 6. **Drag Chute** — Deploy when on runway.
- 7. **Battery Switch** — OFF.

■ LANDING WITH ONE OR BOTH MAIN GEAR UP

If both main gear cannot be extended
with nose gear up or down
and all gear cannot be retracted:

- 1. **Eject.**

If landing must be attempted:

- 1. **Throttles** — IDLE at touchdown.
With one main gear extended, touch down in center of runway. Use aileron to hold wings level.
- 2. **Drag Chute** — Deploy.
- 3. **Throttles** — OFF.
- 4. **Battery Switch** — OFF.

LANDING WITH TIRE FAILURE ^D

30

- Fly a normal traffic pattern.
- Land beyond the approach-end arresting cables.

■ NOSE GEAR

- Expend excess fuel, fire out ammunition, and jettison centerline store to obtain a more favorable aft CG position.
- After touchdown, hold nosewheel off runway as long as possible. When nosewheel touches down, deploy drag chute.
- Nosewheel steering angle must not exceed 20 degrees left or right.

■ MAIN GEAR

- Braking efficiency on the ground is reduced.
- Expend excess fuel and jettison stores, but retain empty pylon fuel tanks.
- Land on the side of the runway away from the failed tire. When it is unknown which tire is failed, touch down in the center of the runway.
- After touchdown, deploy drag chute.

NO-FLAP LANDING &
FLAP ASYMMETRY

31

■ NO-FLAP LANDING

When landing with the wing flaps retracted, use the normal landing procedure modified as follows:

- 1. **Pattern** — Fly wider than normal.
- 2. **Final turn, final approach, and touchdown airspeeds** — Increase by 10 kts.
- 3. **AOA Indicator** — 16.4 units on final approach.
Landing distance increases approximately 15%.

■ FLAP ASYMMETRY

When landing with a flap asymmetry, including one flap damaged or lost:

- 1. **Controllability check** — Perform /13.
Use UP or FULL flaps. Do not change flap position during approach and go-around.
- 2. **Follow the No-Flap Landing procedure.**

